

BFS Traversal

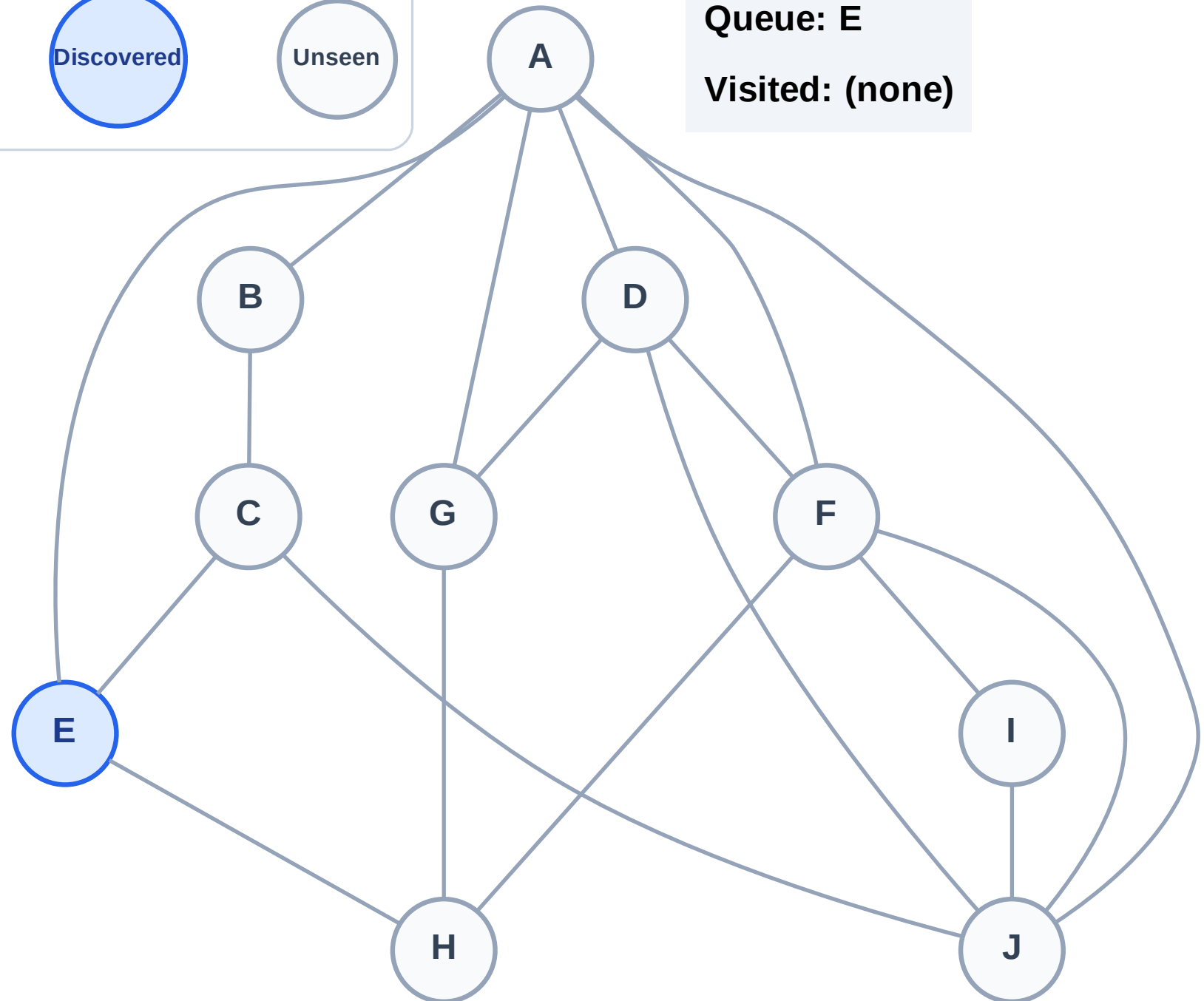
Step 1: Start at E

Legend



Queue: E

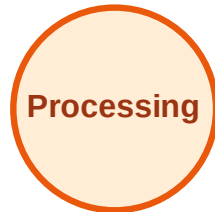
Visited: (none)



BFS Traversal

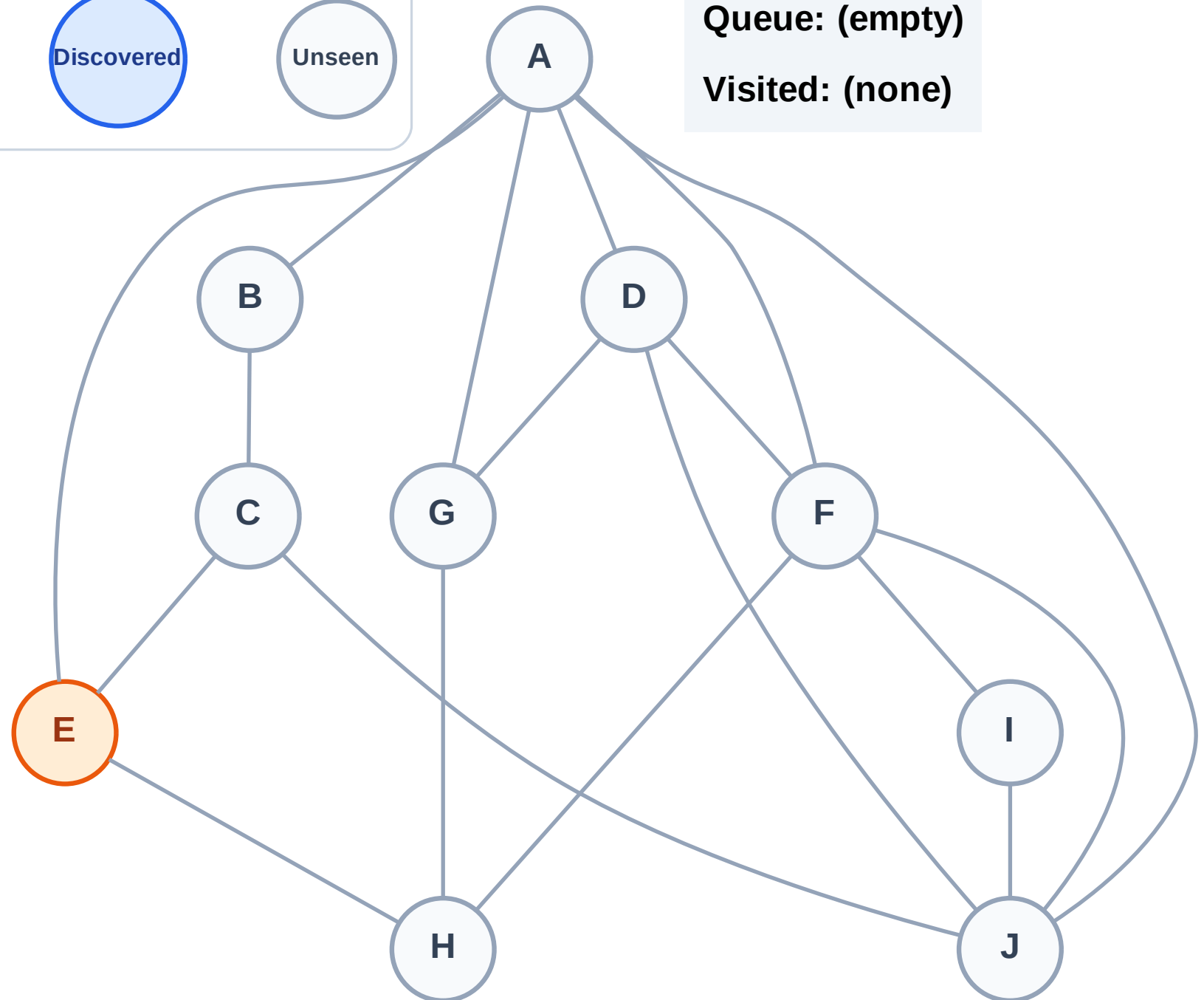
Step 2: Process node E

Legend



Queue: (empty)

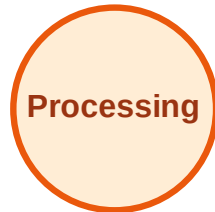
Visited: (none)



BFS Traversal

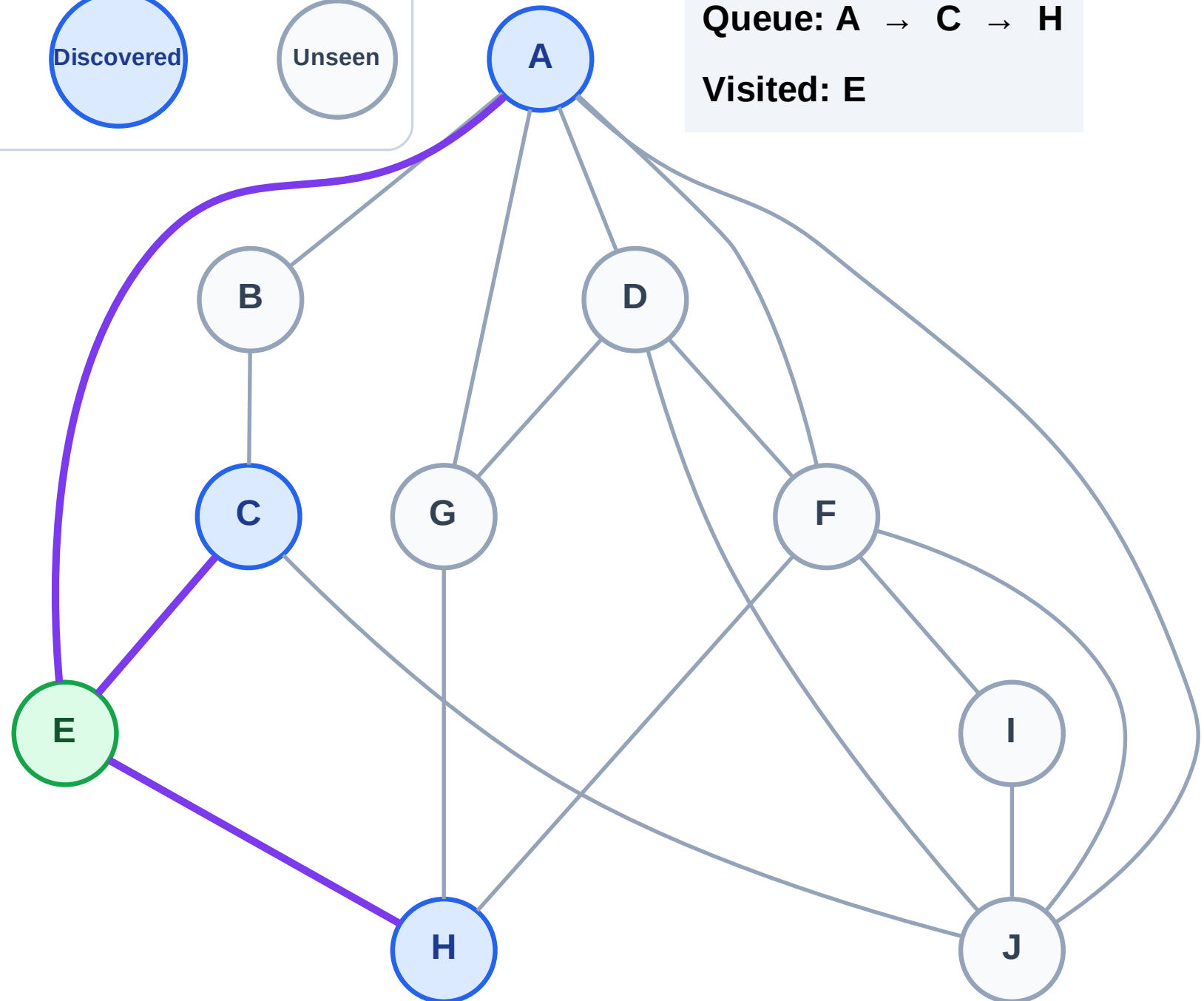
Step 3: Discover A, C, H from E

Legend



Queue: A → C → H

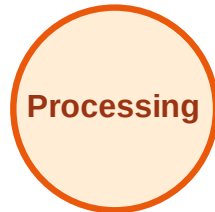
Visited: E



BFS Traversal

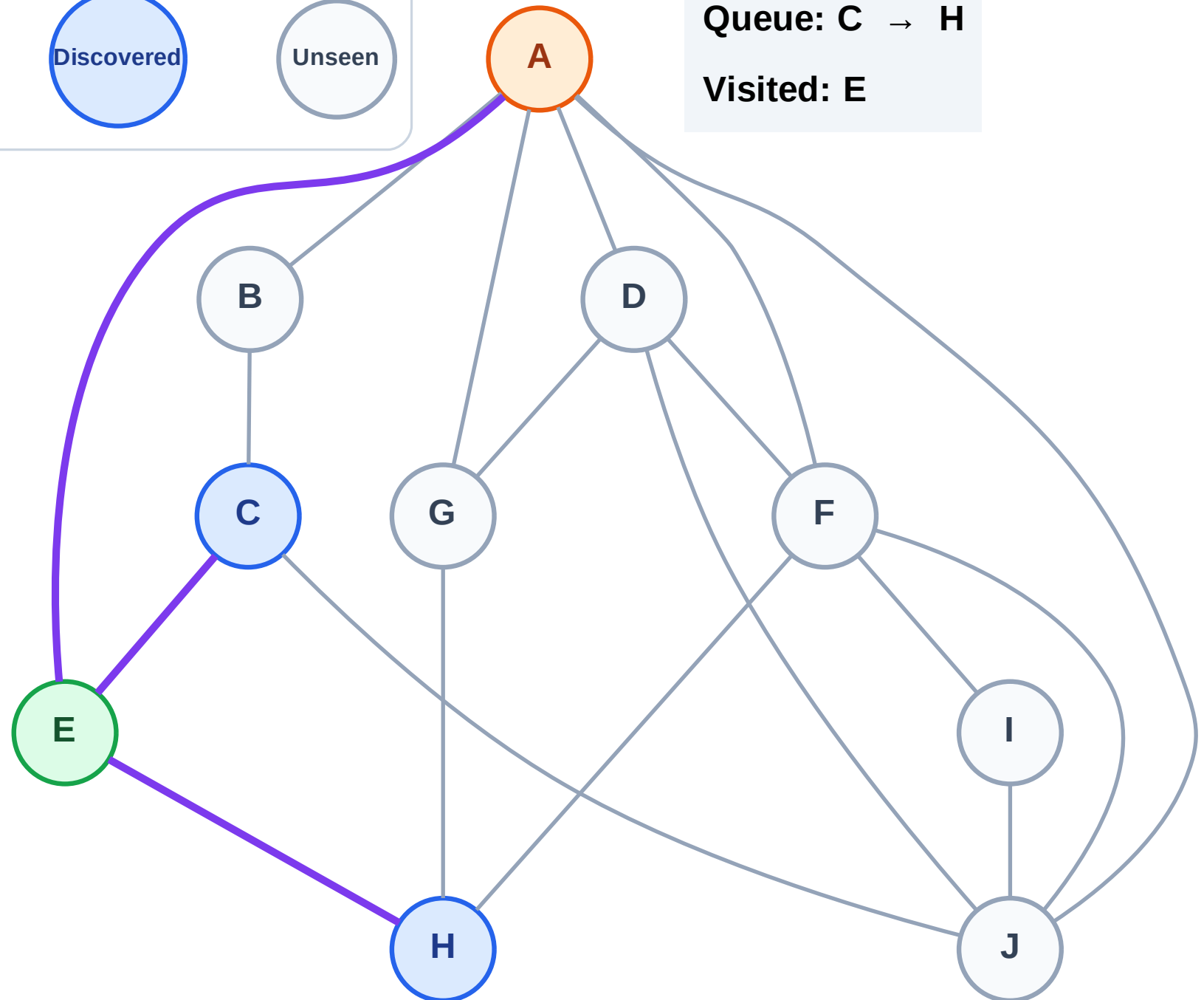
Step 4: Process node A

Legend



Queue: C → H

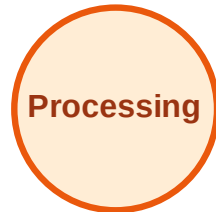
Visited: E



BFS Traversal

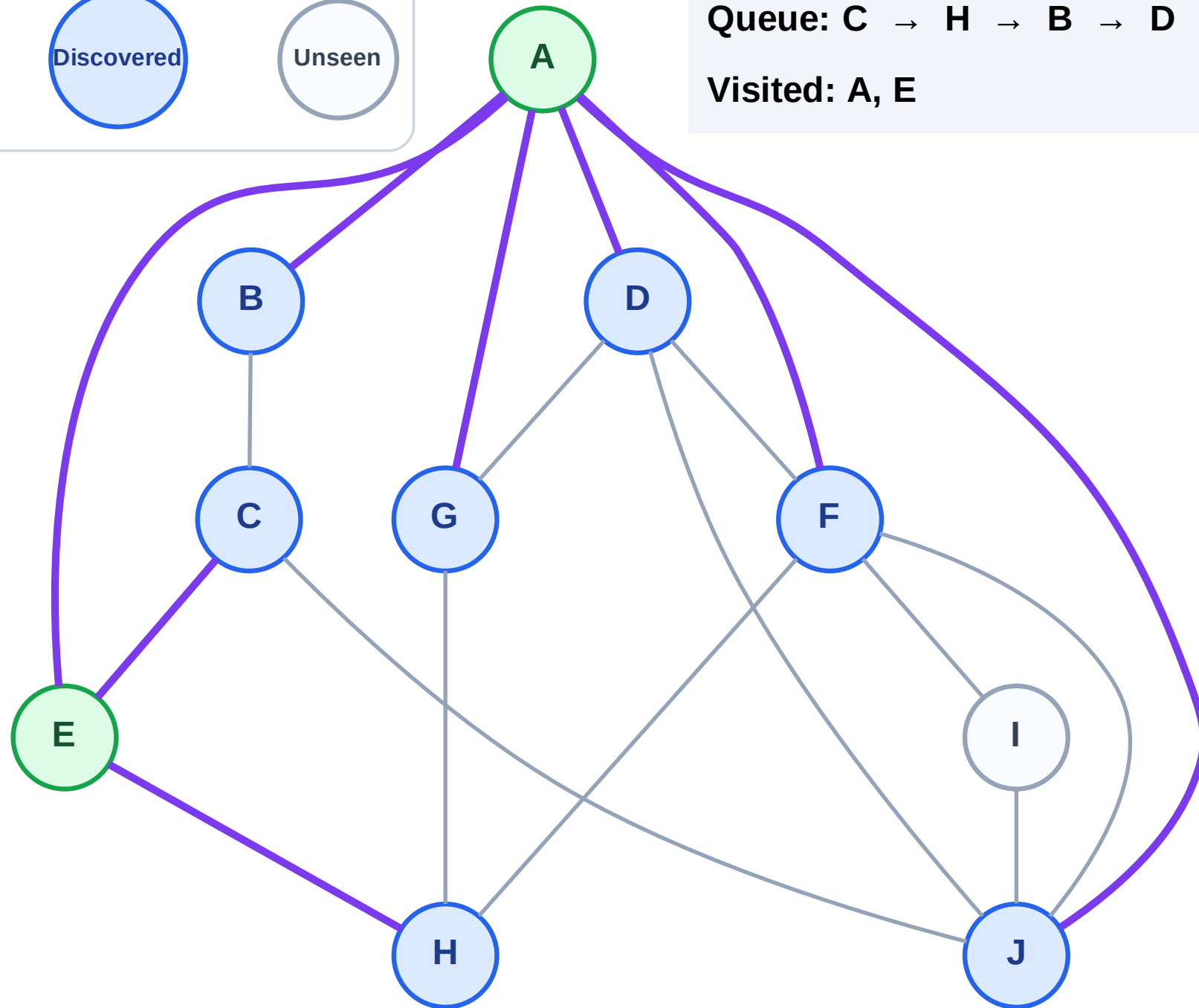
Step 5: Discover B, D, F, G, J from A

Legend



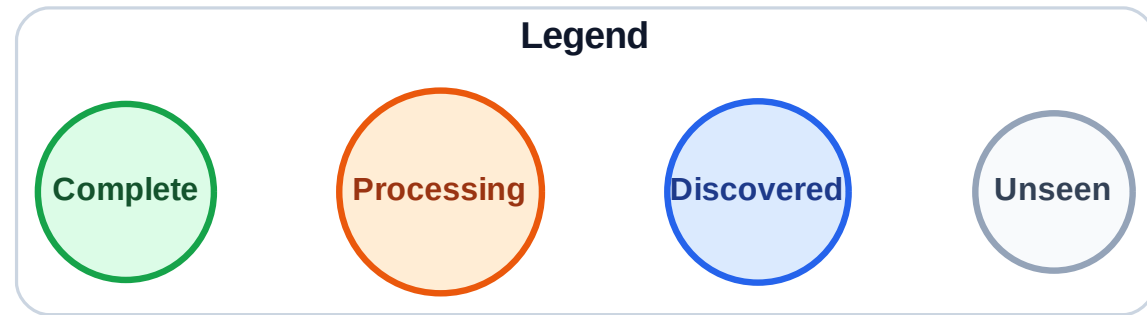
Queue: C → H → B → D → F → G → J

Visited: A, E



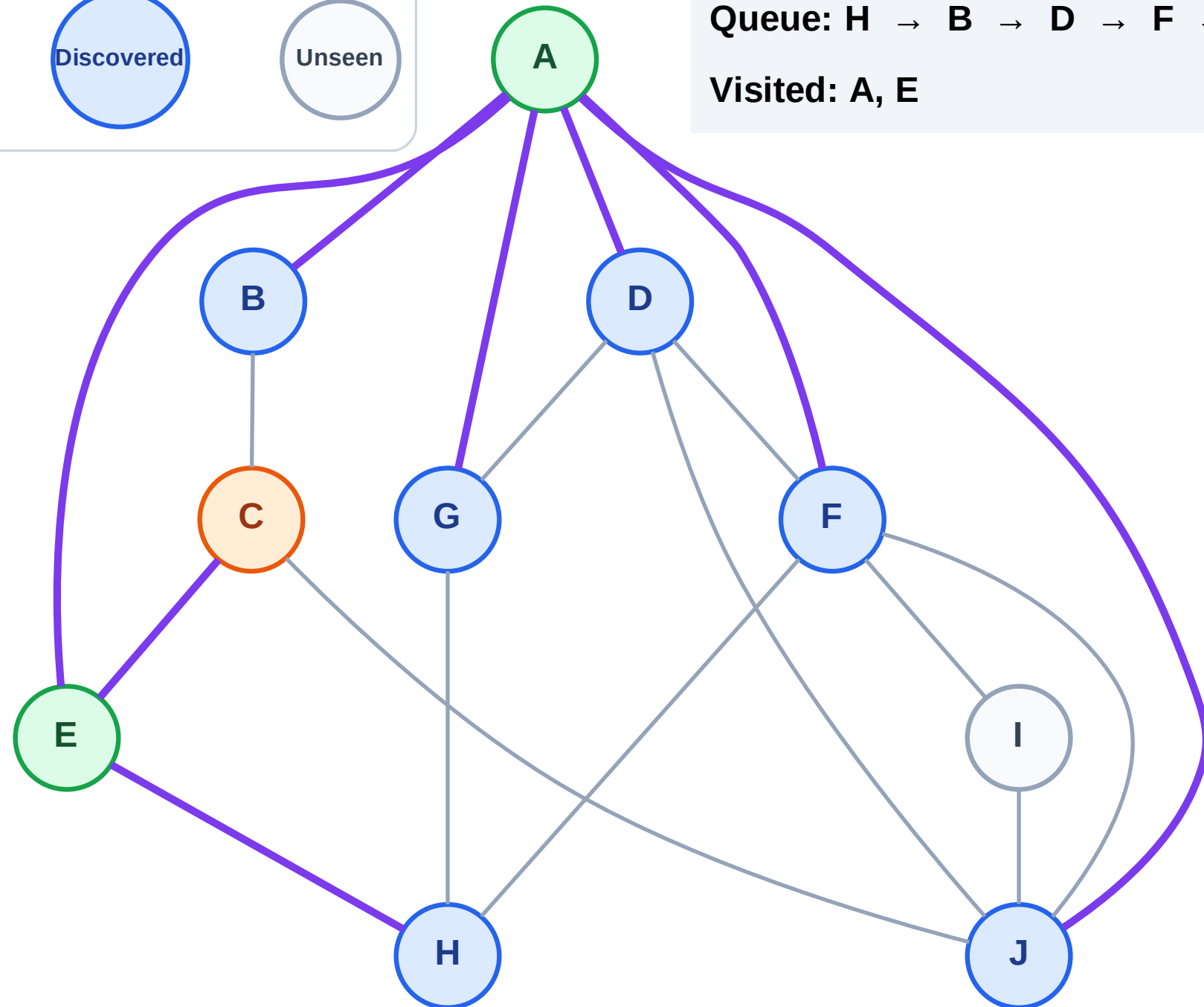
BFS Traversal

Step 6: Process node C



Queue: H → B → D → F → G → J

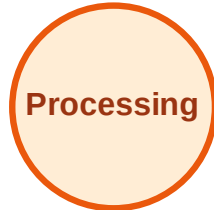
Visited: A, E



BFS Traversal

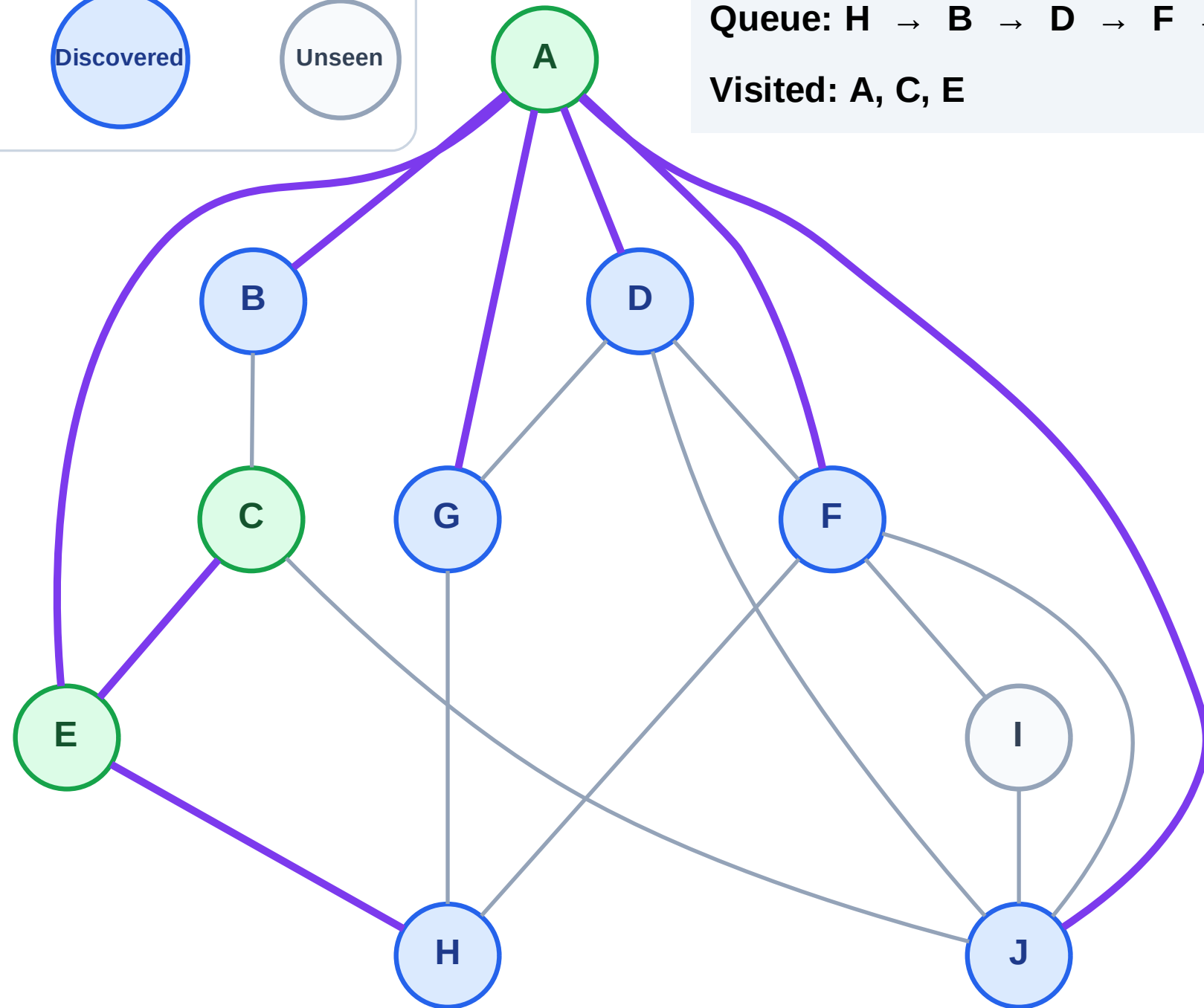
Step 7: No new neighbors from C

Legend



Queue: H → B → D → F → G → J

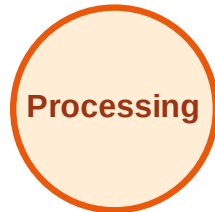
Visited: A, C, E



BFS Traversal

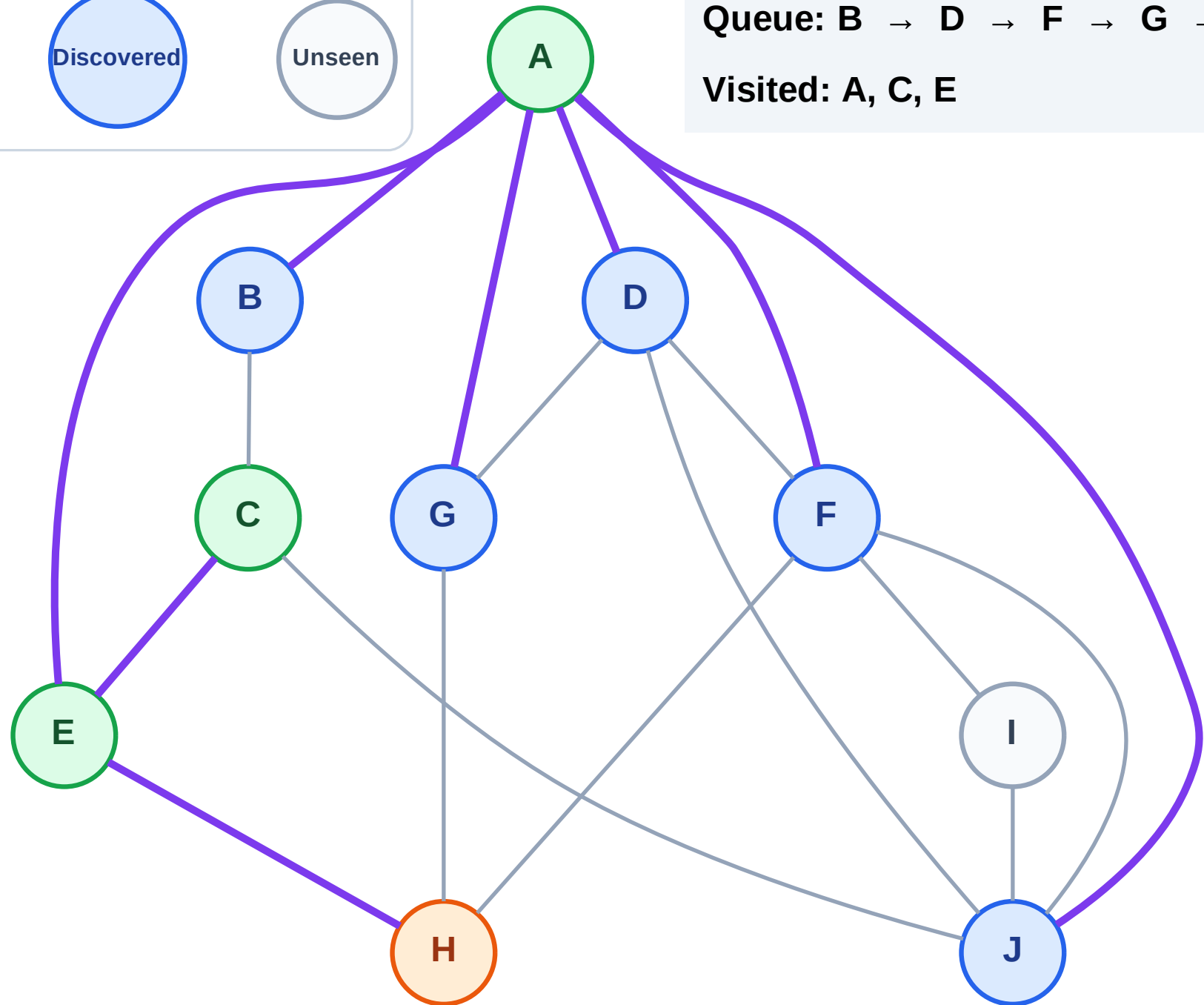
Step 8: Process node H

Legend



Queue: B → D → F → G → J

Visited: A, C, E



BFS Traversal

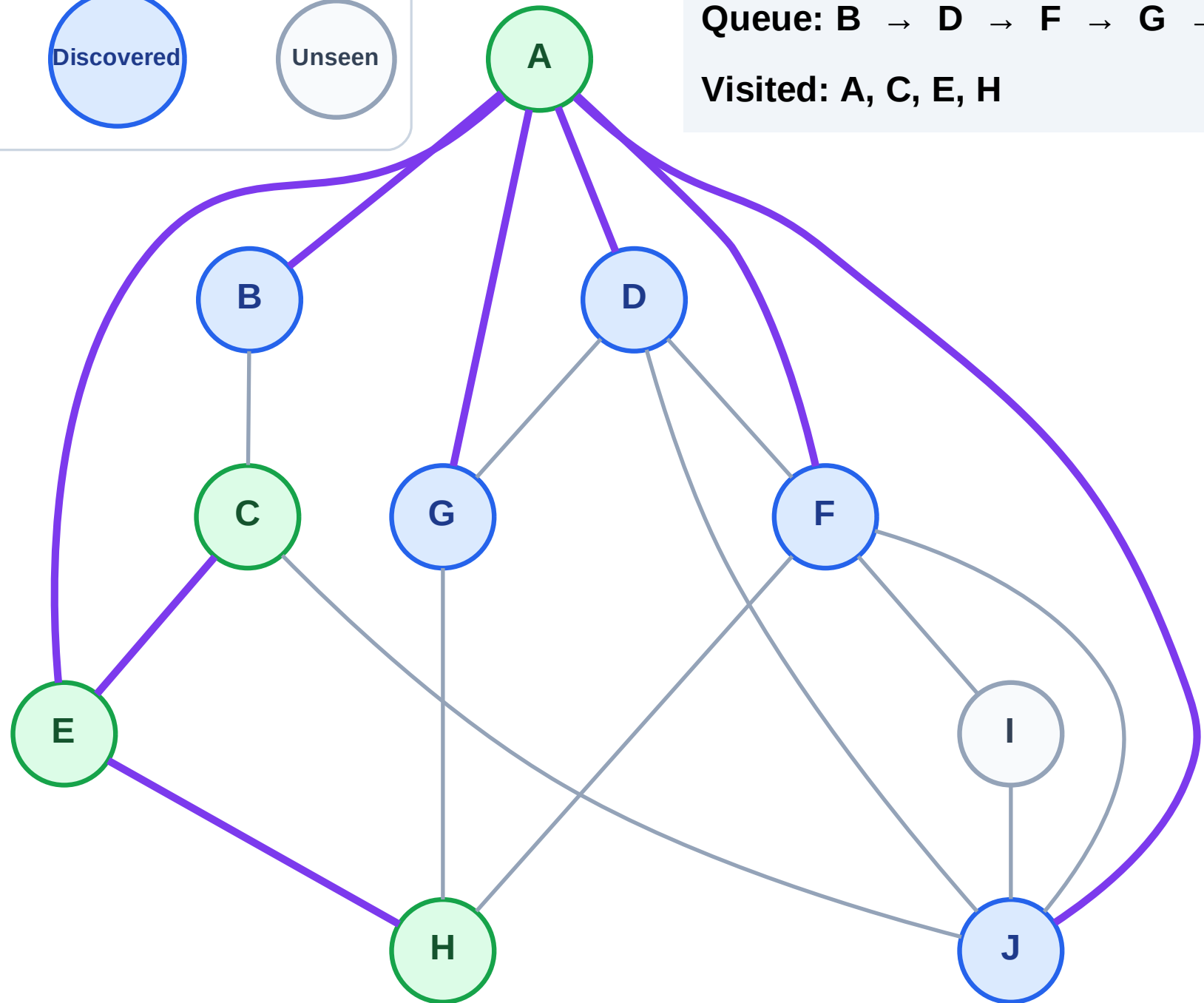
Step 9: No new neighbors from H

Legend



Queue: B → D → F → G → J

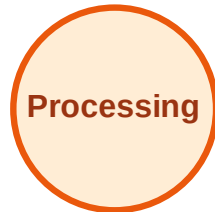
Visited: A, C, E, H



BFS Traversal

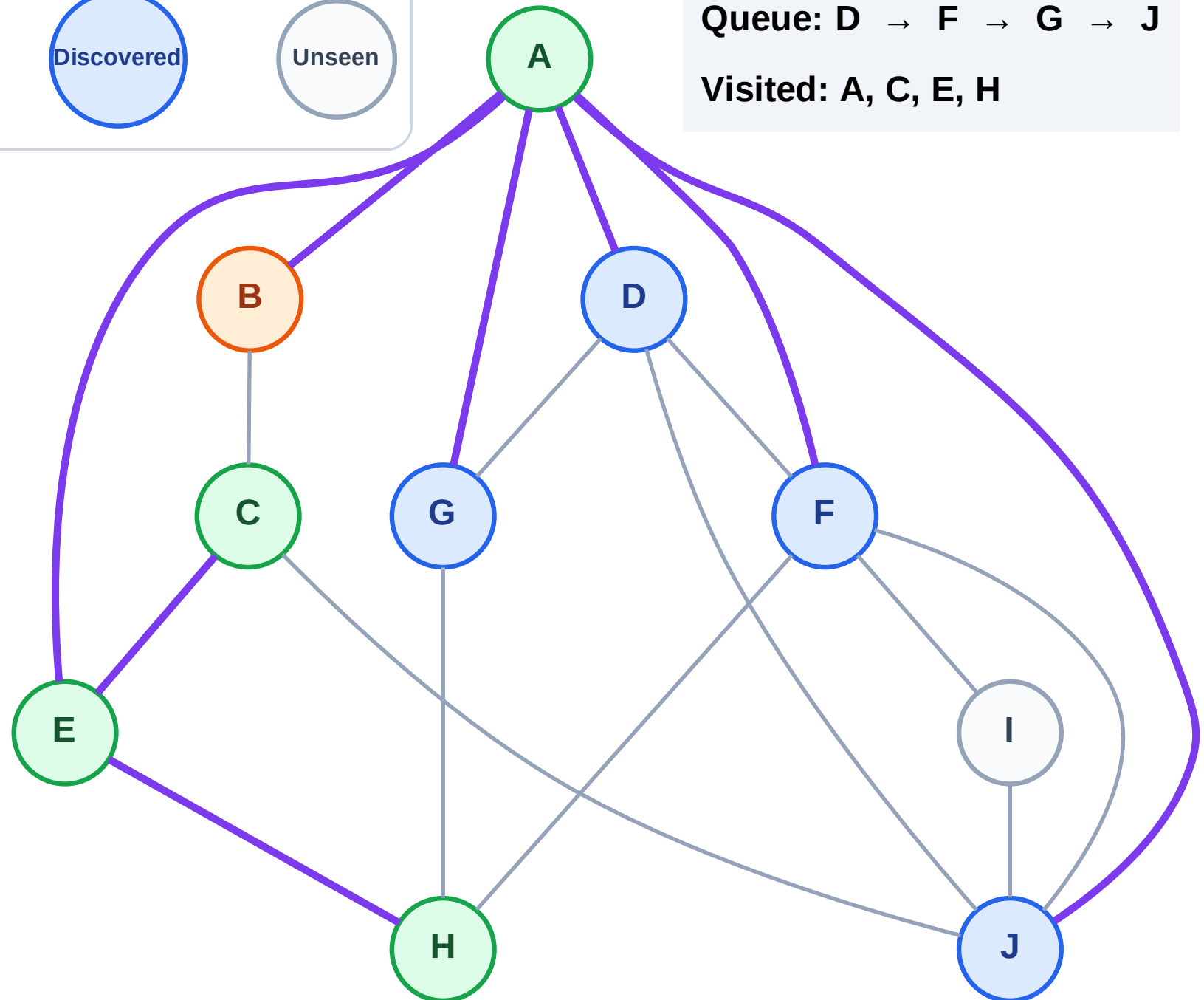
Step 10: Process node B

Legend



Queue: D → F → G → J

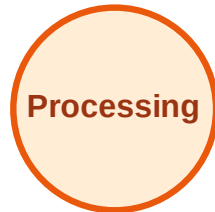
Visited: A, C, E, H



BFS Traversal

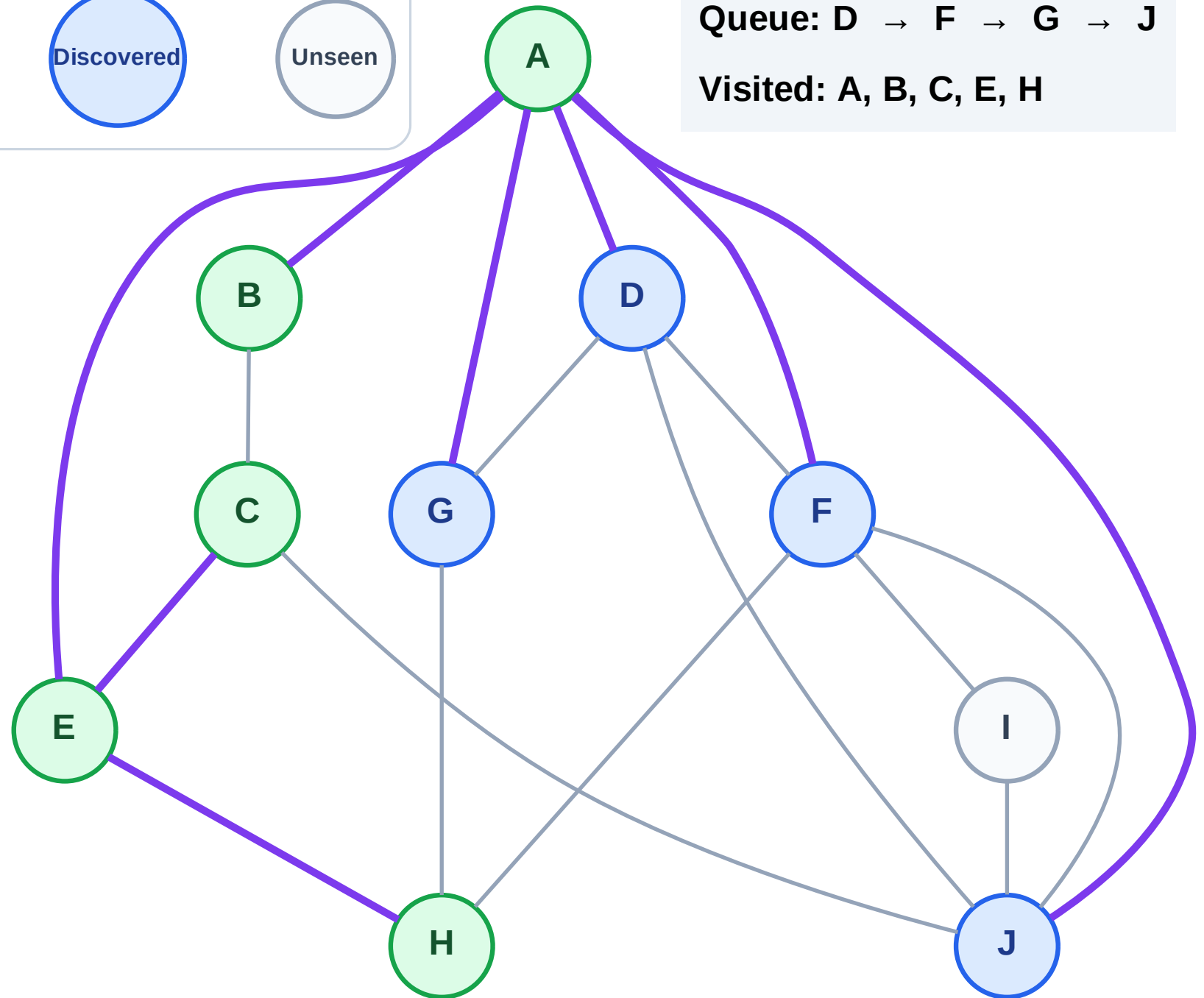
Step 11: No new neighbors from B

Legend



Queue: D → F → G → J

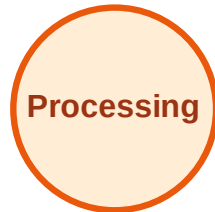
Visited: A, B, C, E, H



BFS Traversal

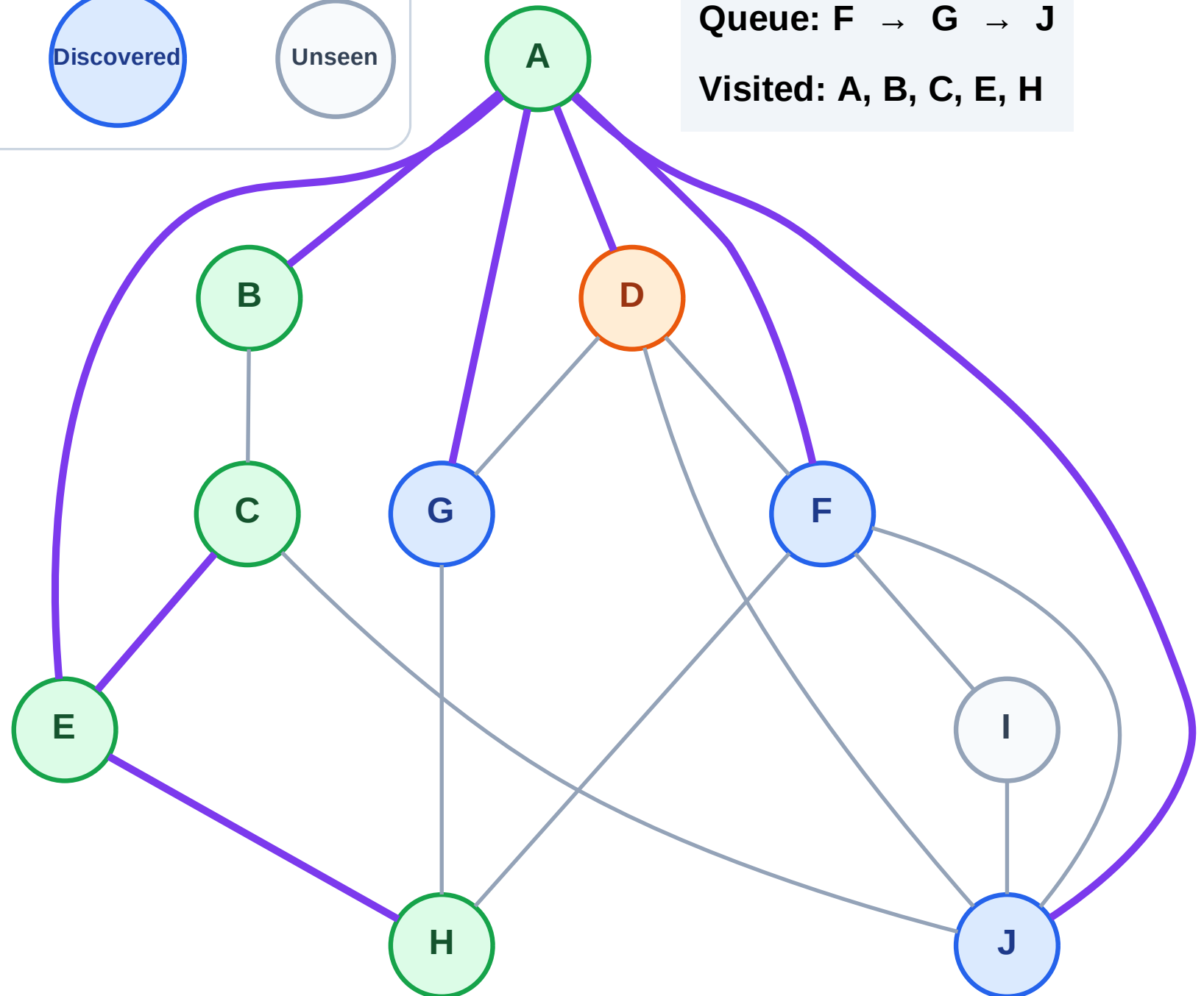
Step 12: Process node D

Legend



Queue: F → G → J

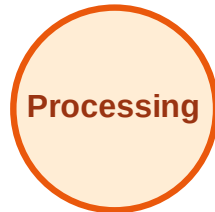
Visited: A, B, C, E, H



BFS Traversal

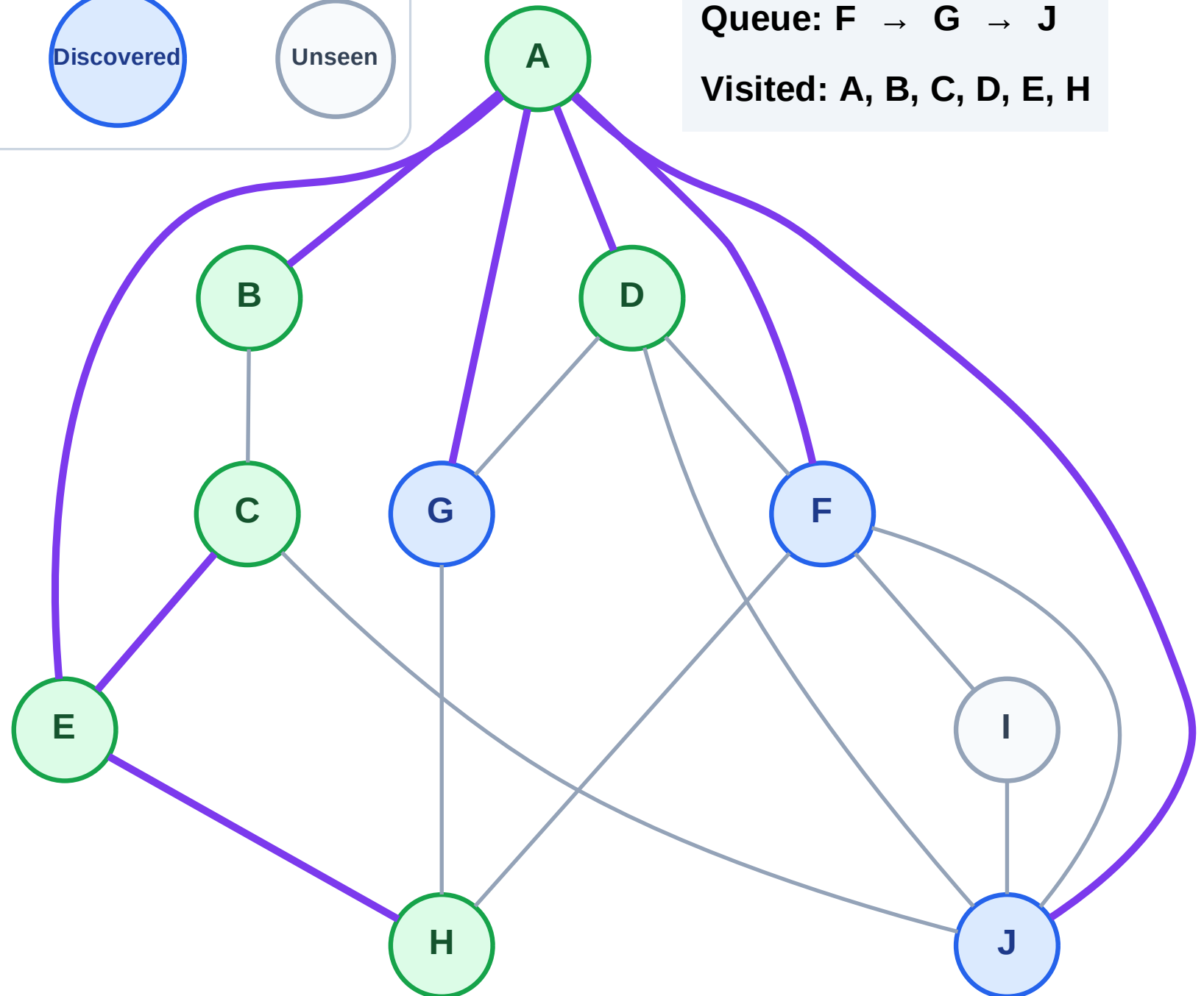
Step 13: No new neighbors from D

Legend



Queue: F → G → J

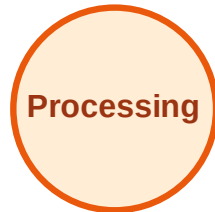
Visited: A, B, C, D, E, H



BFS Traversal

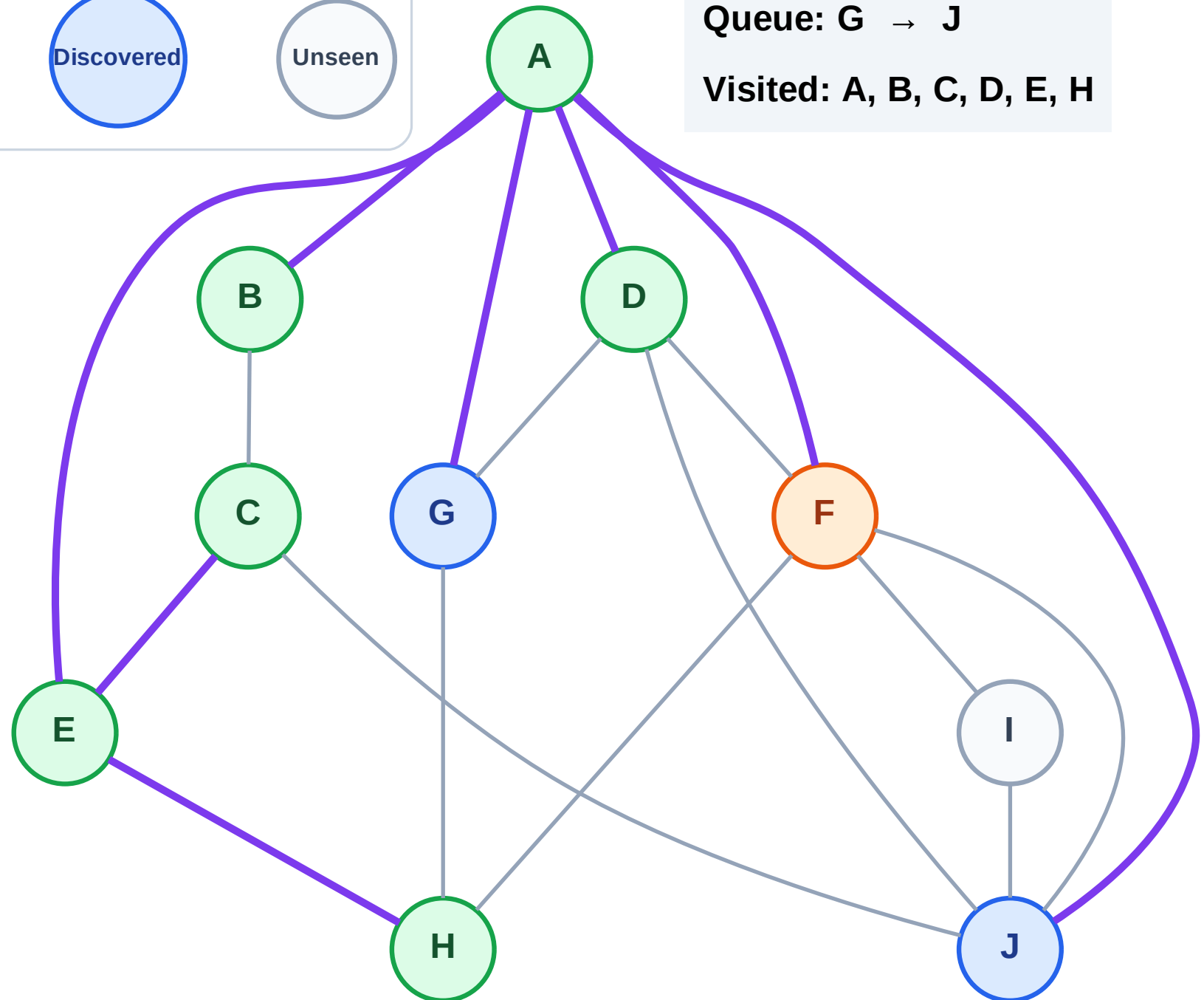
Step 14: Process node F

Legend



Queue: G → J

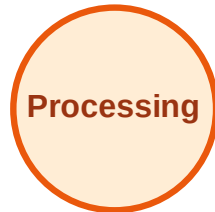
Visited: A, B, C, D, E, H



BFS Traversal

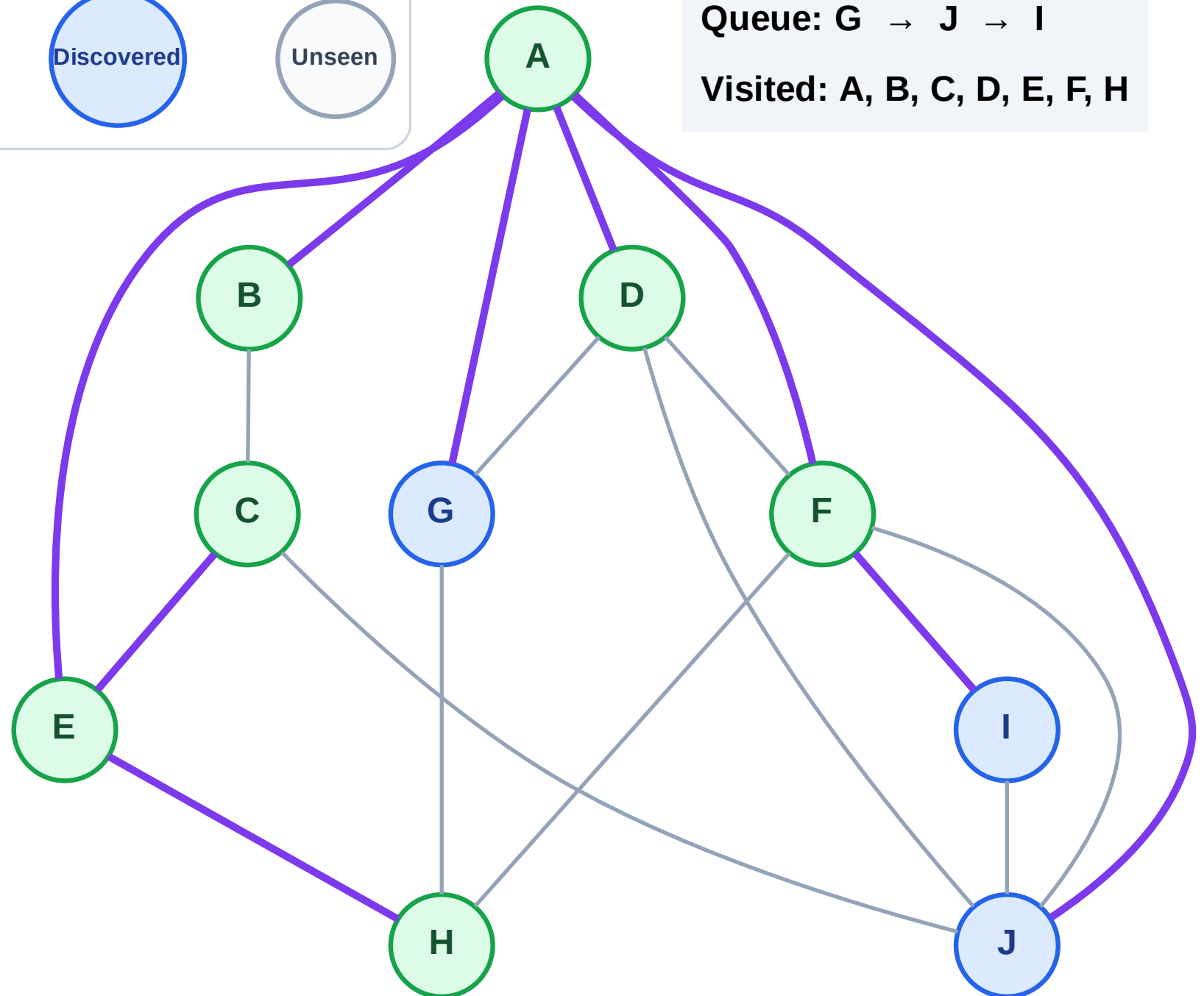
Step 15: Discover I from F

Legend



Queue: G → J → I

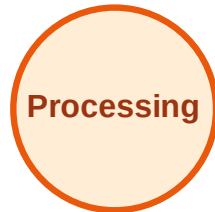
Visited: A, B, C, D, E, F, H



BFS Traversal

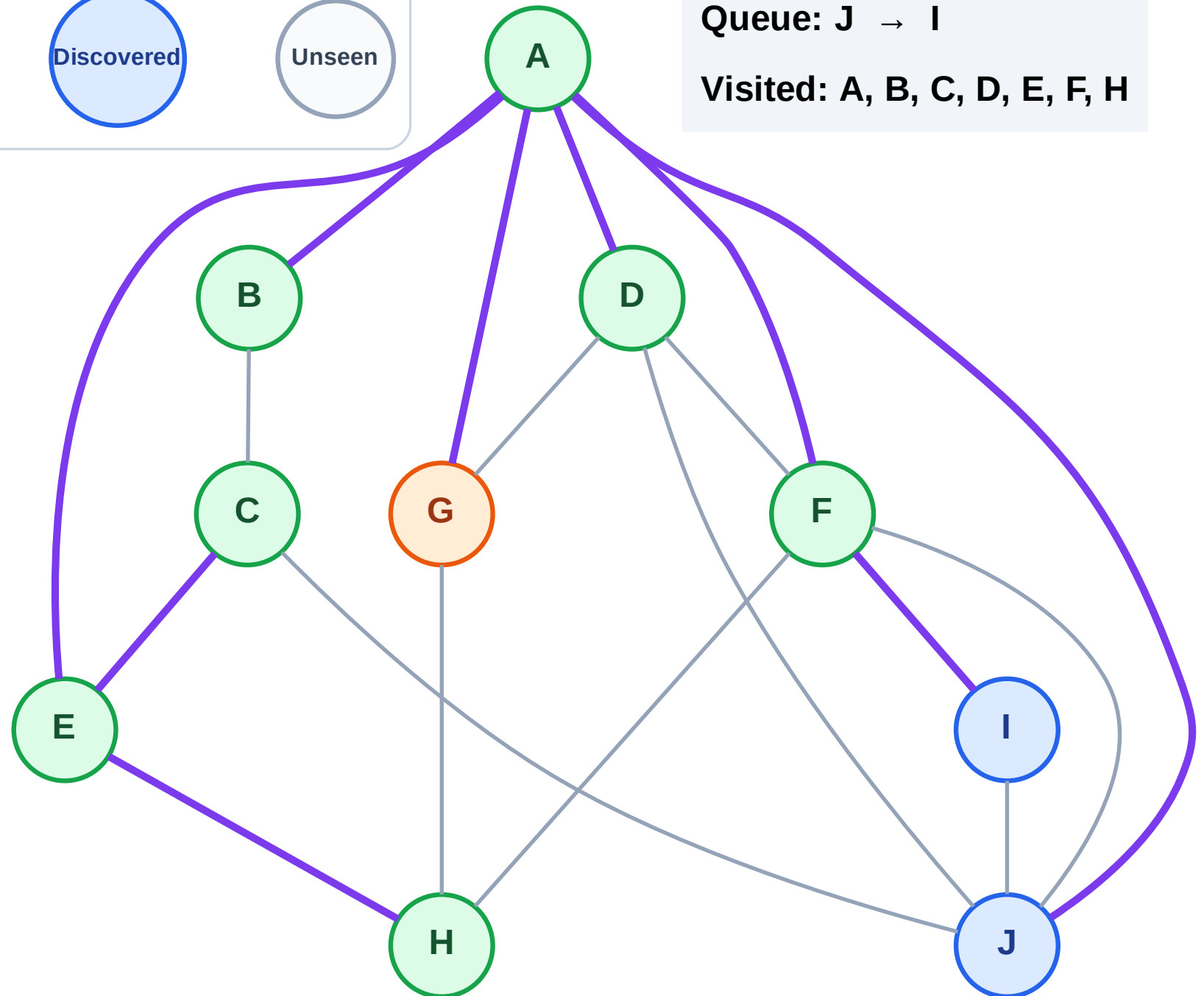
Step 16: Process node G

Legend



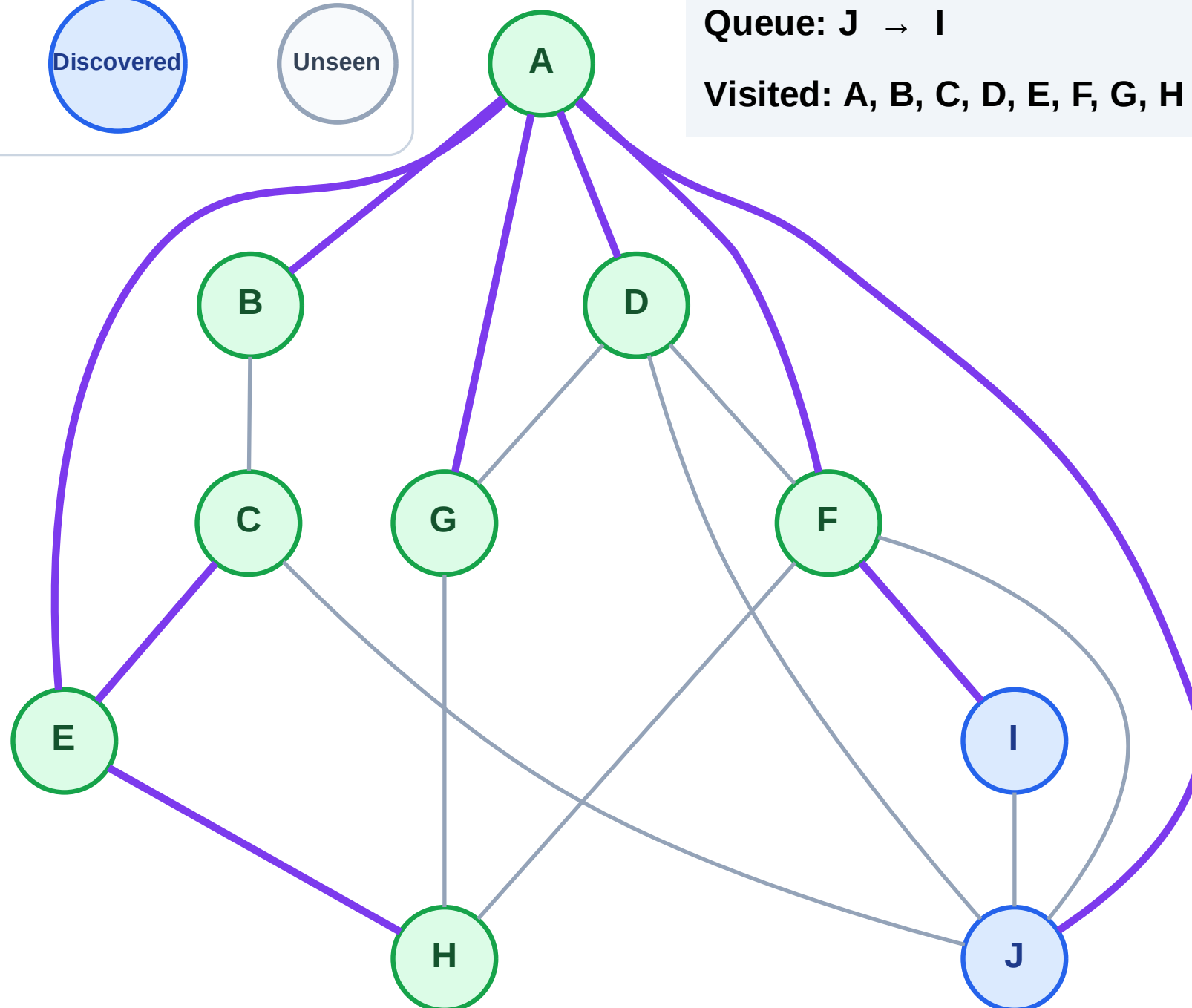
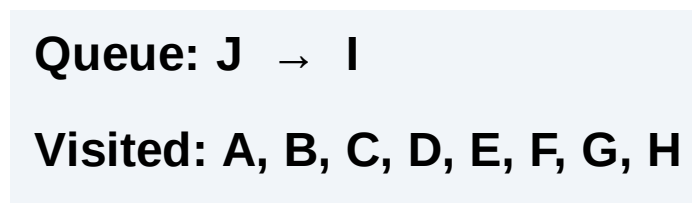
Queue: J → I

Visited: A, B, C, D, E, F, H



Step 17: No new neighbors from G

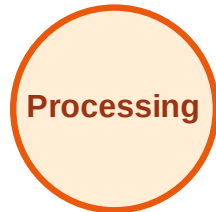
Step 17: No new neighbors from G



BFS Traversal

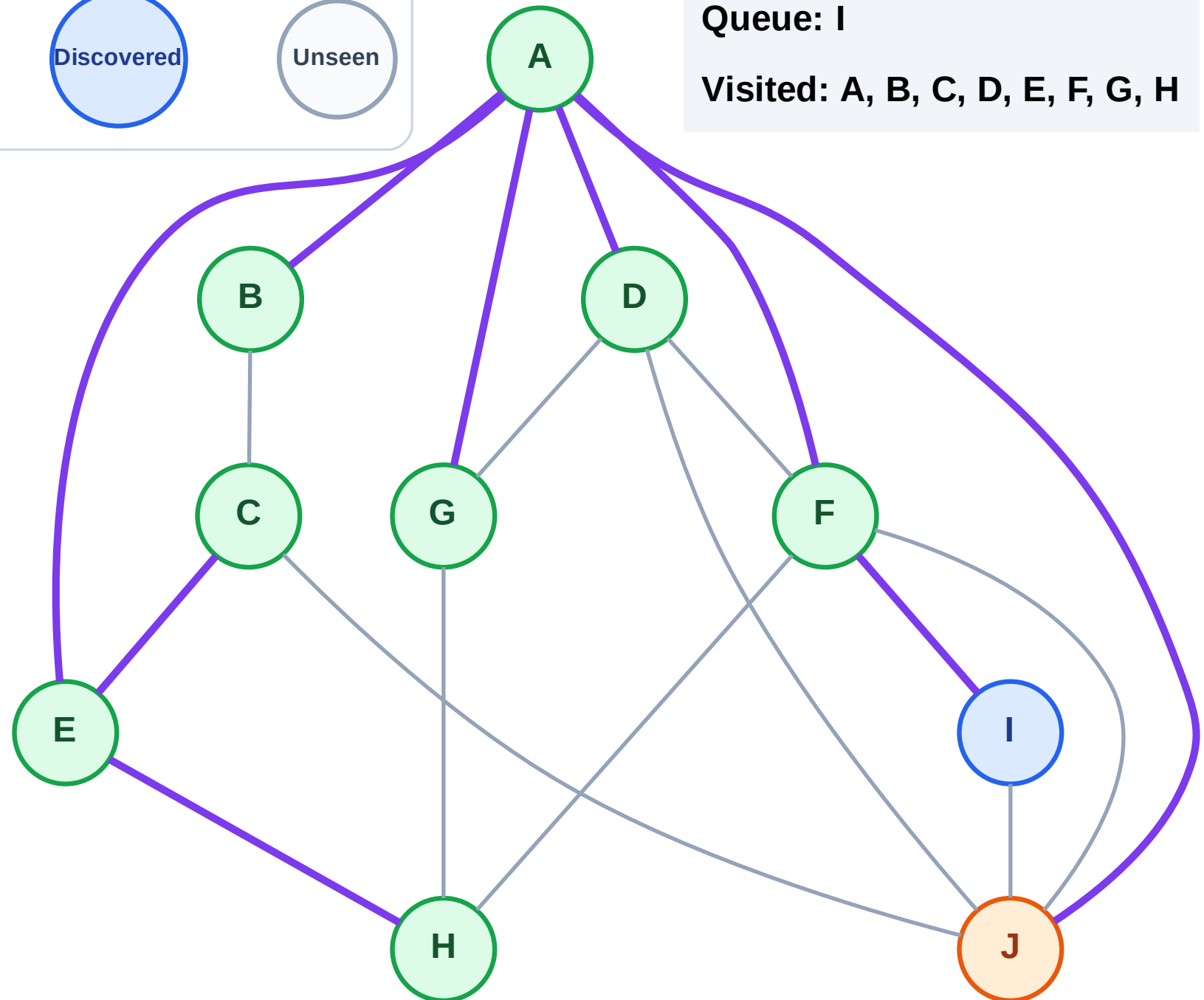
Step 18: Process node J

Legend



Queue: I

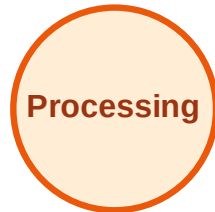
Visited: A, B, C, D, E, F, G, H



BFS Traversal

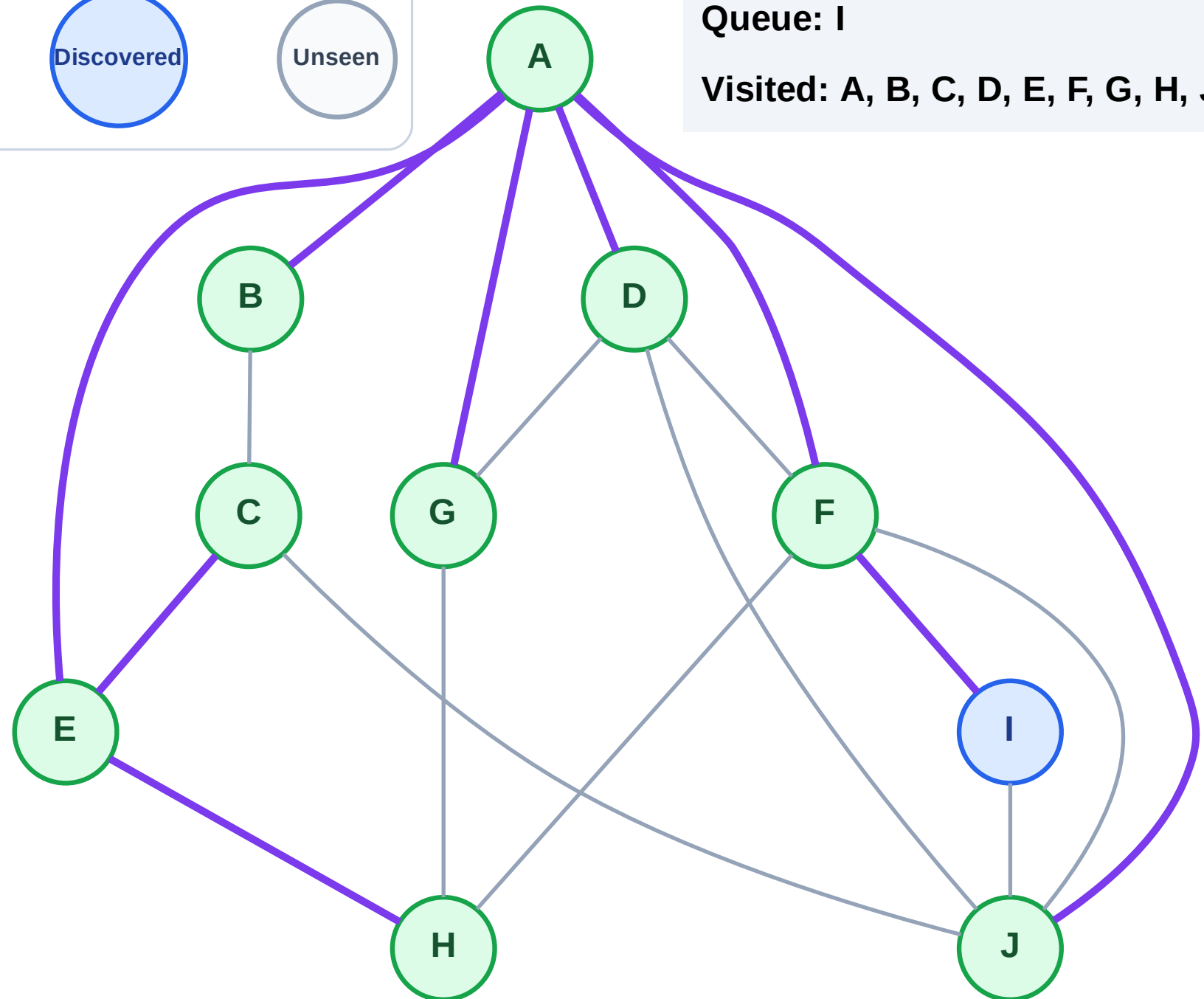
Step 19: No new neighbors from J

Legend



Queue: I

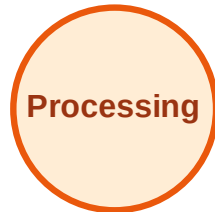
Visited: A, B, C, D, E, F, G, H, J



BFS Traversal

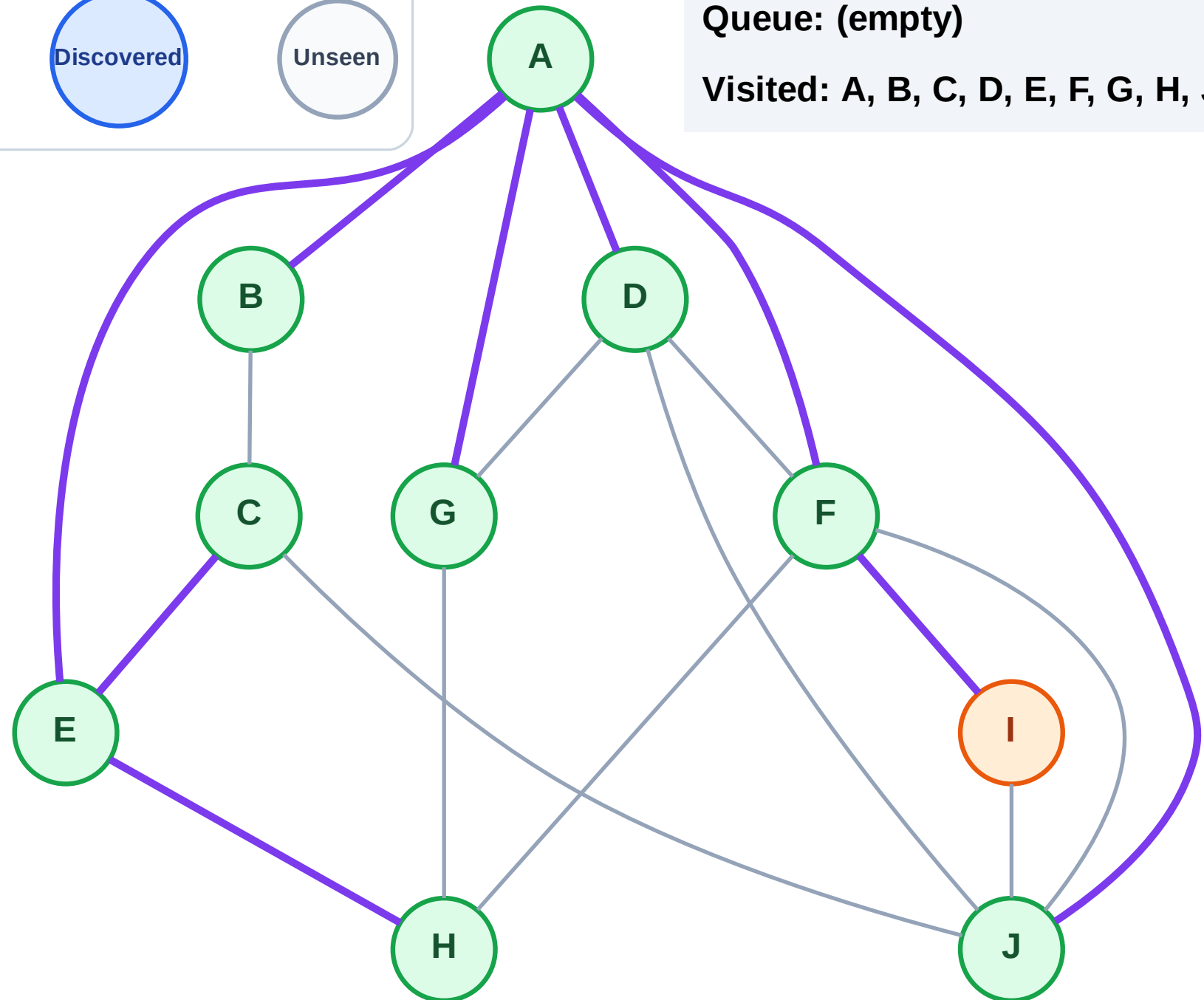
Step 20: Process node I

Legend



Queue: (empty)

Visited: A, B, C, D, E, F, G, H, J



BFS Traversal

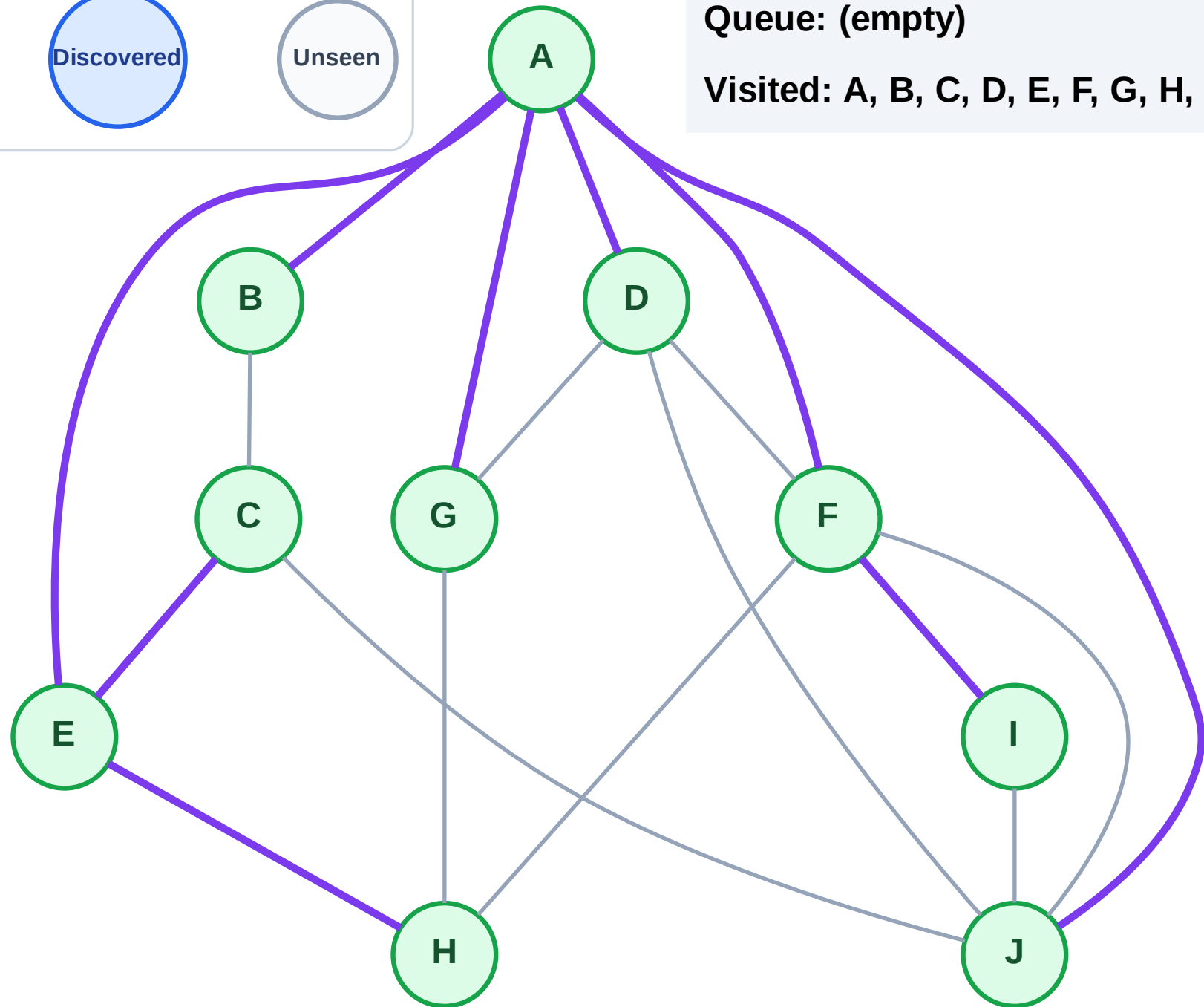
Step 21: No new neighbors from I

Legend



Queue: (empty)

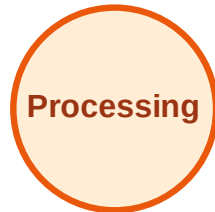
Visited: A, B, C, D, E, F, G, H, I, J



DFS Traversal

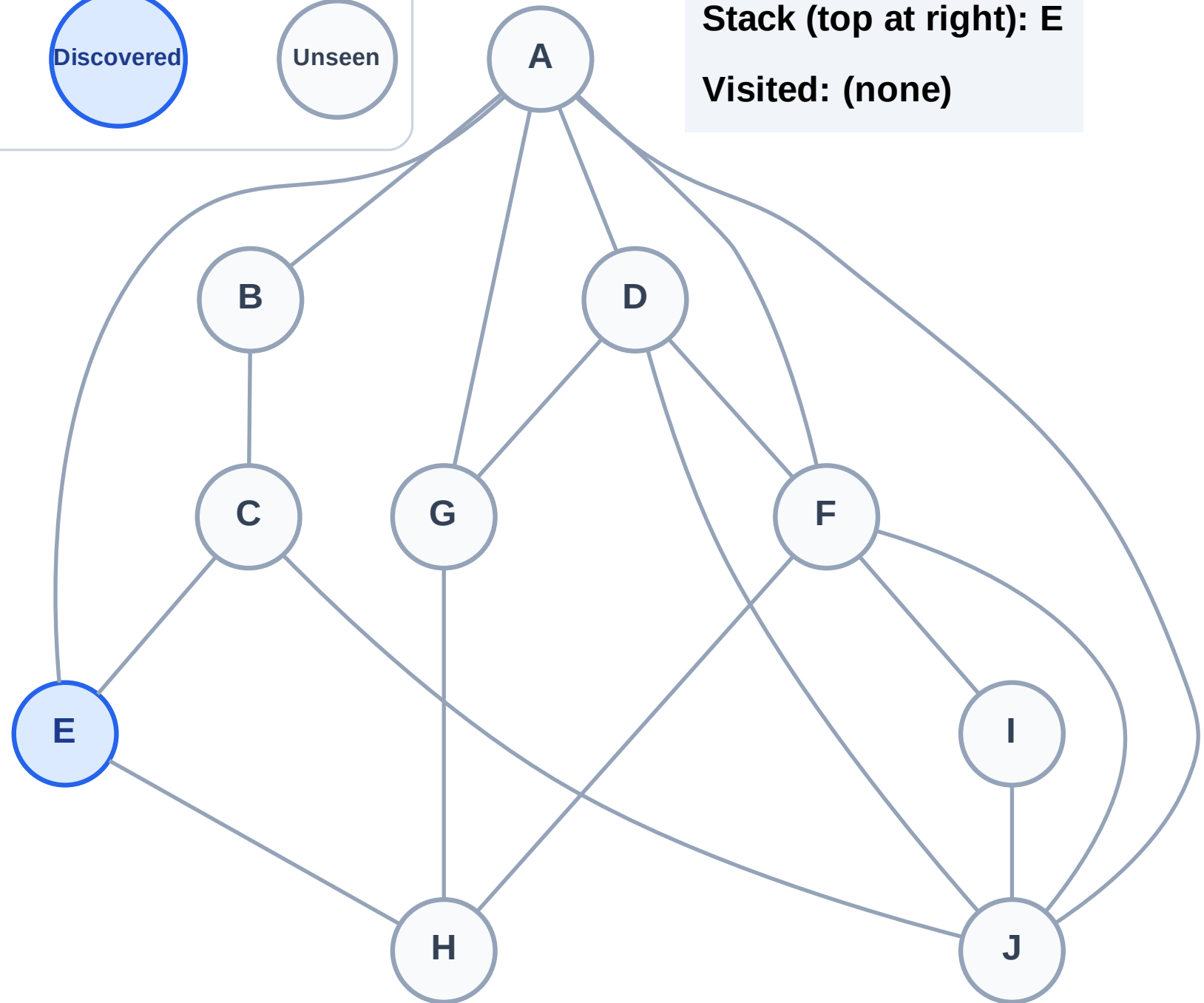
Step 1: Start at E

Legend



Stack (top at right): E

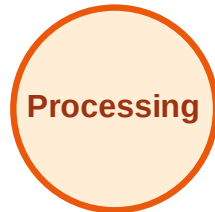
Visited: (none)



DFS Traversal

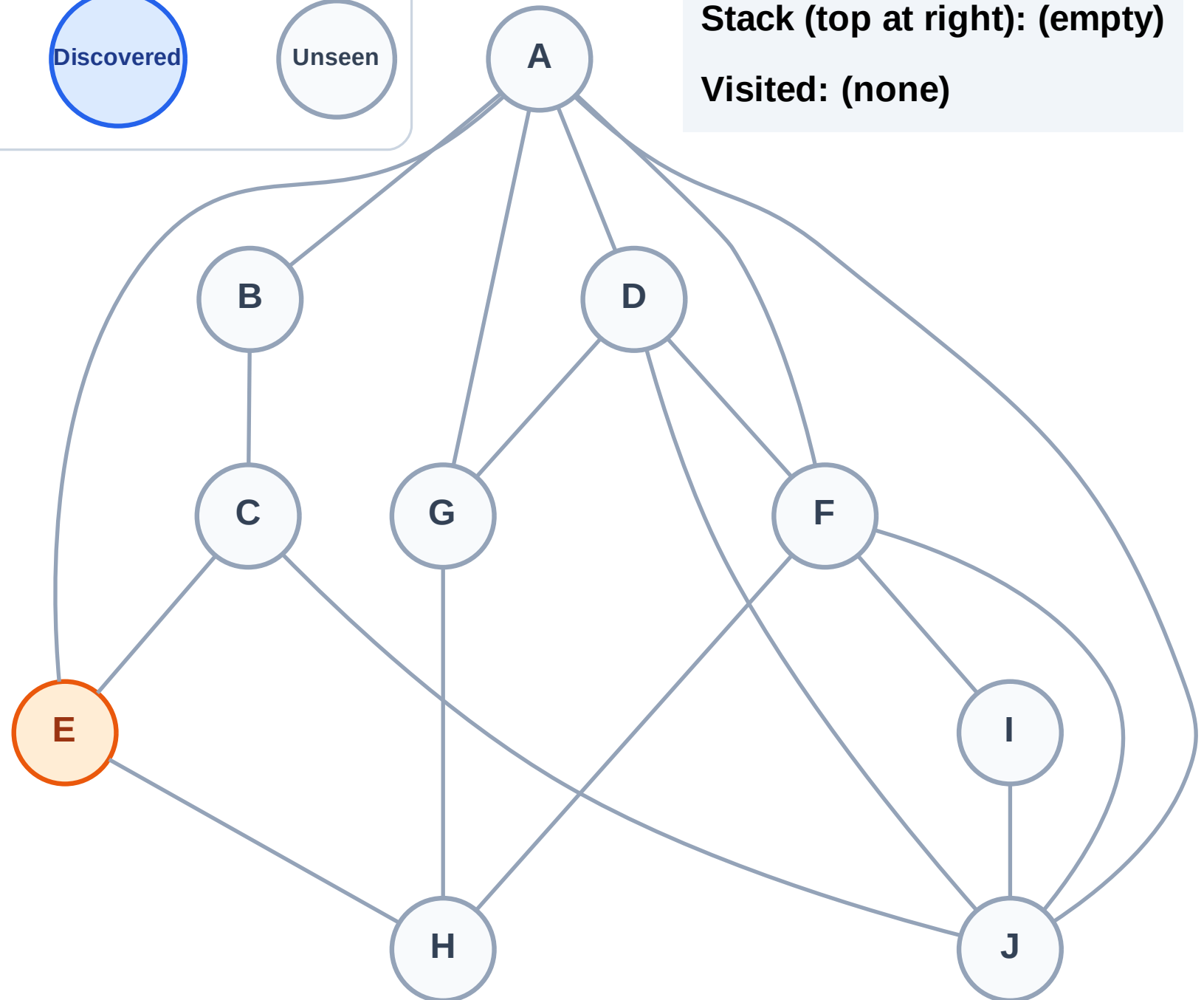
Step 2: Process node E

Legend



Stack (top at right): (empty)

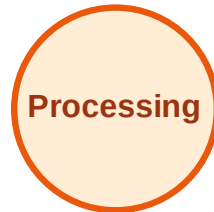
Visited: (none)



DFS Traversal

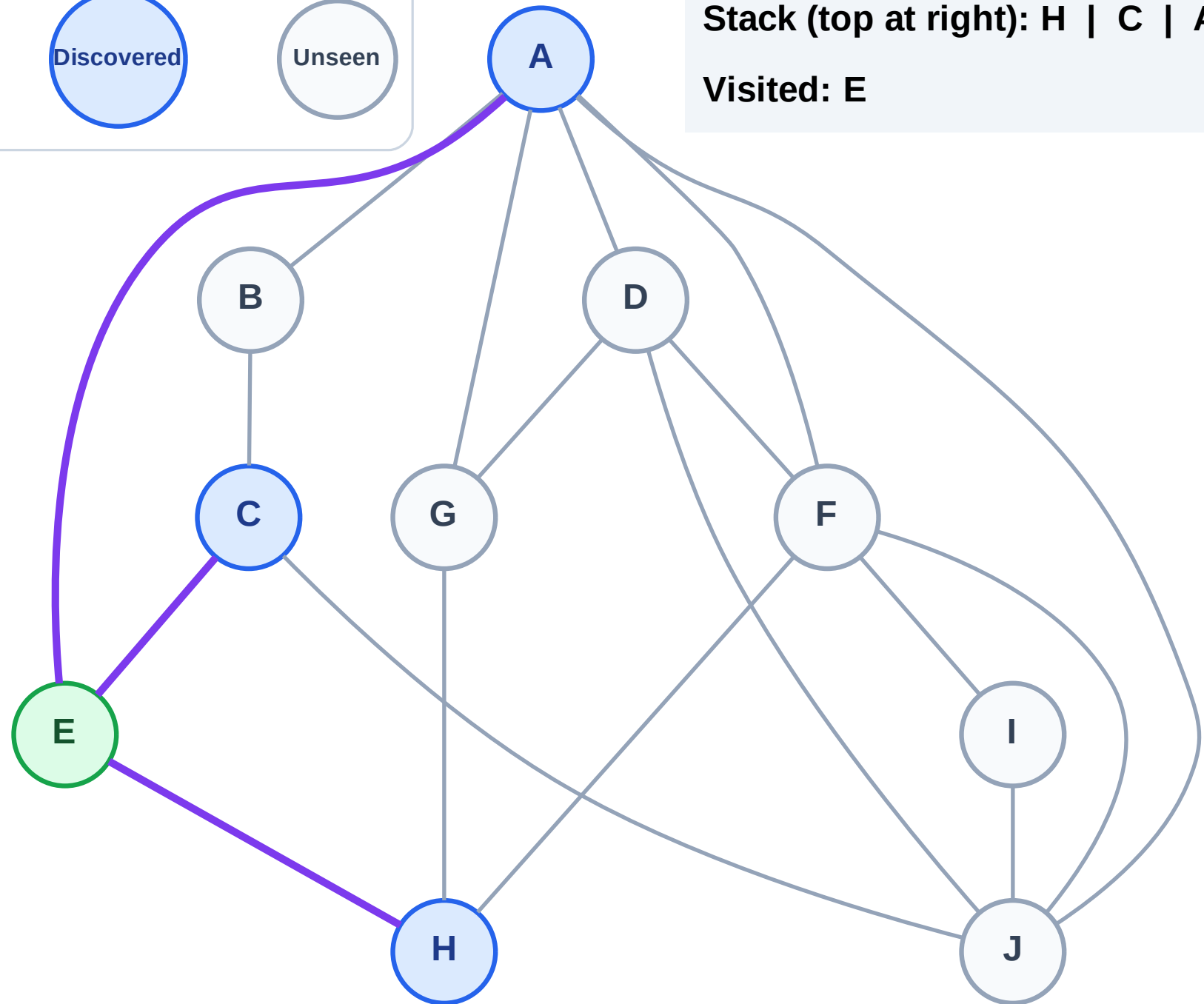
Step 3: Discover H, C, A from E

Legend



Stack (top at right): H | C | A

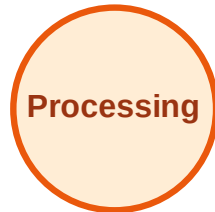
Visited: E



DFS Traversal

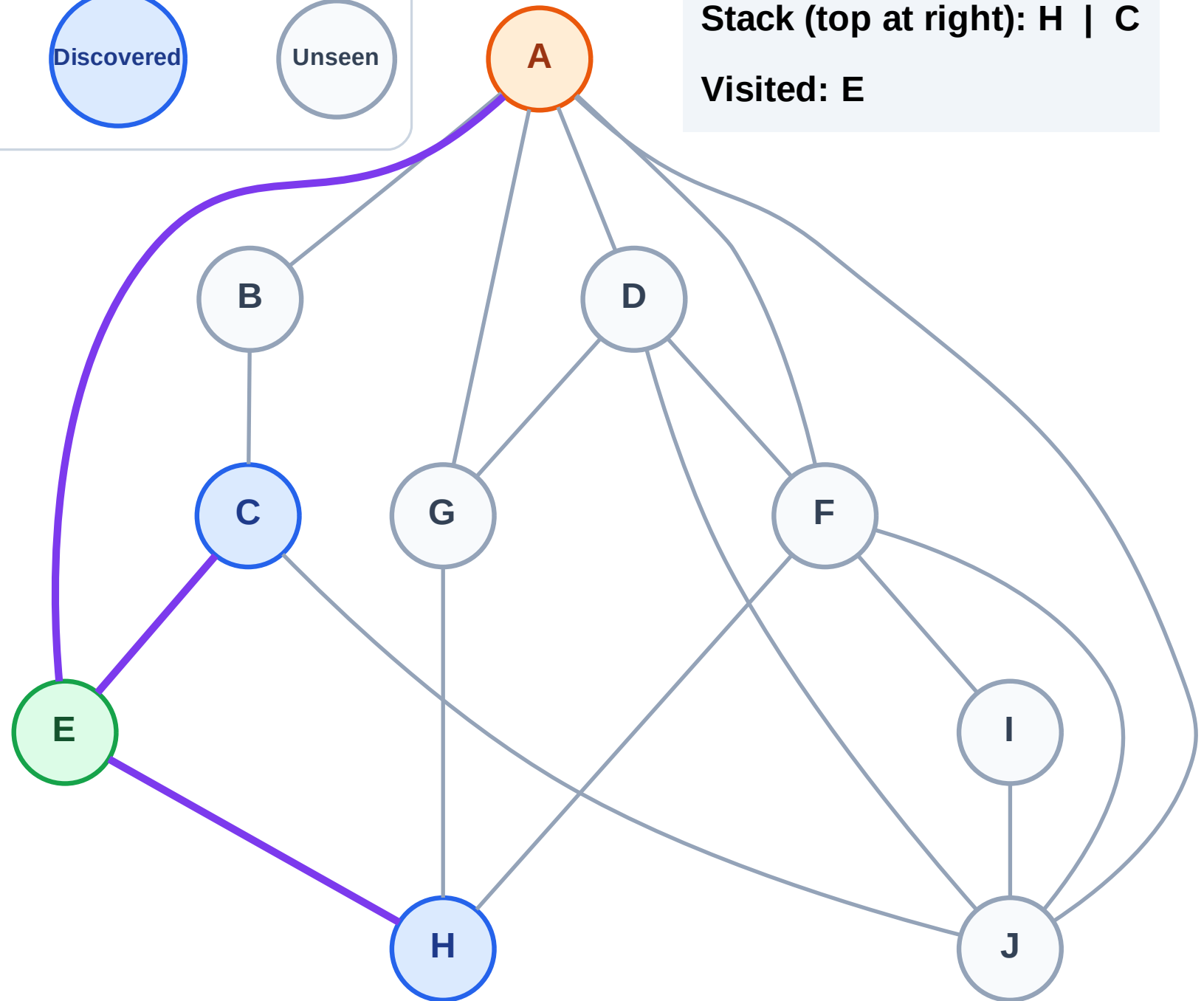
Step 4: Process node A

Legend



Stack (top at right): H | C

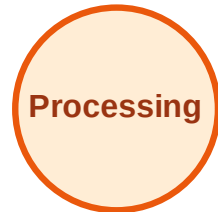
Visited: E



DFS Traversal

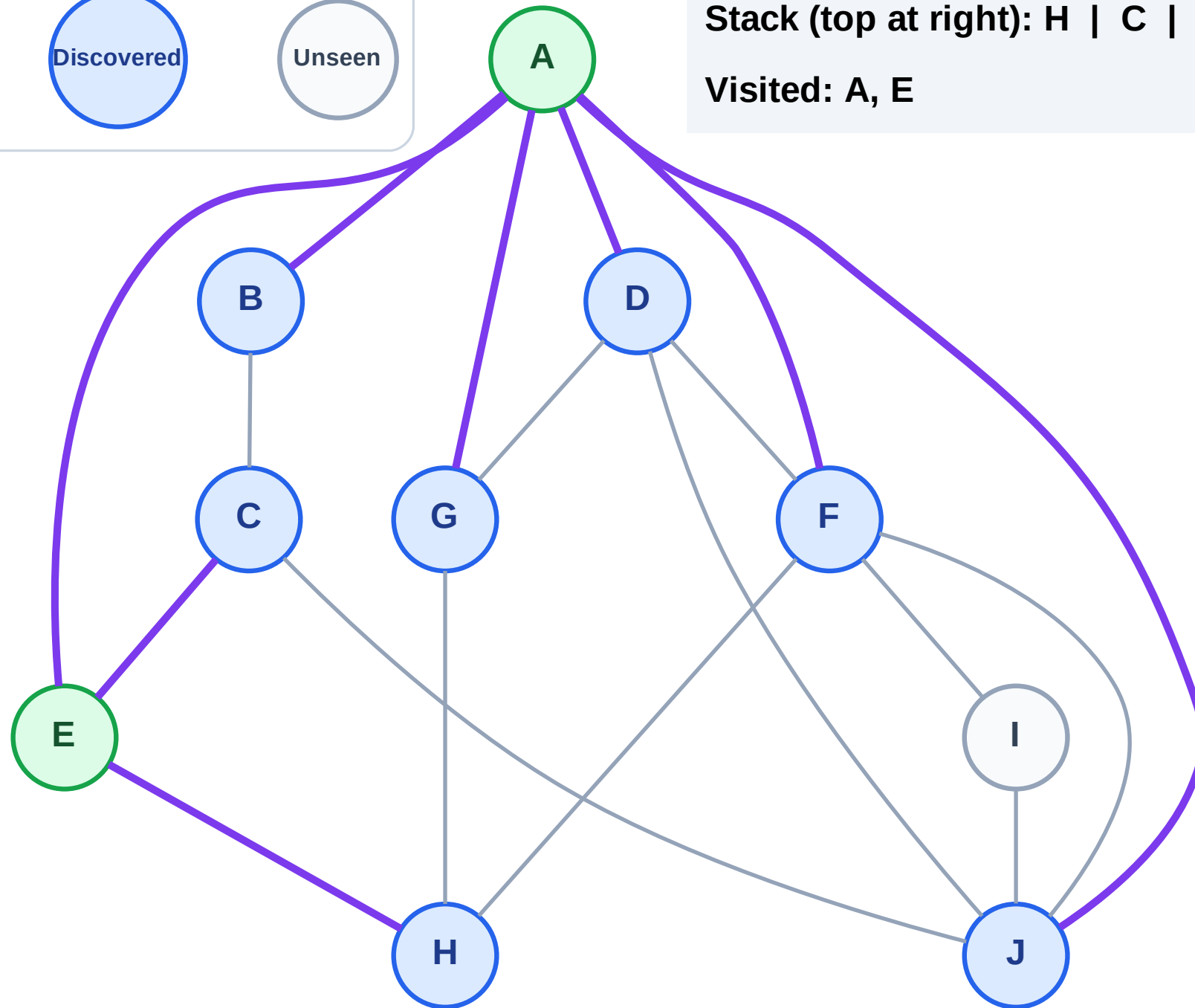
Step 5: Discover J, G, F, D, B from A

Legend



Stack (top at right): H | C | J | G | F | D | B

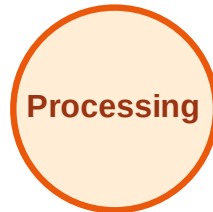
Visited: A, E



DFS Traversal

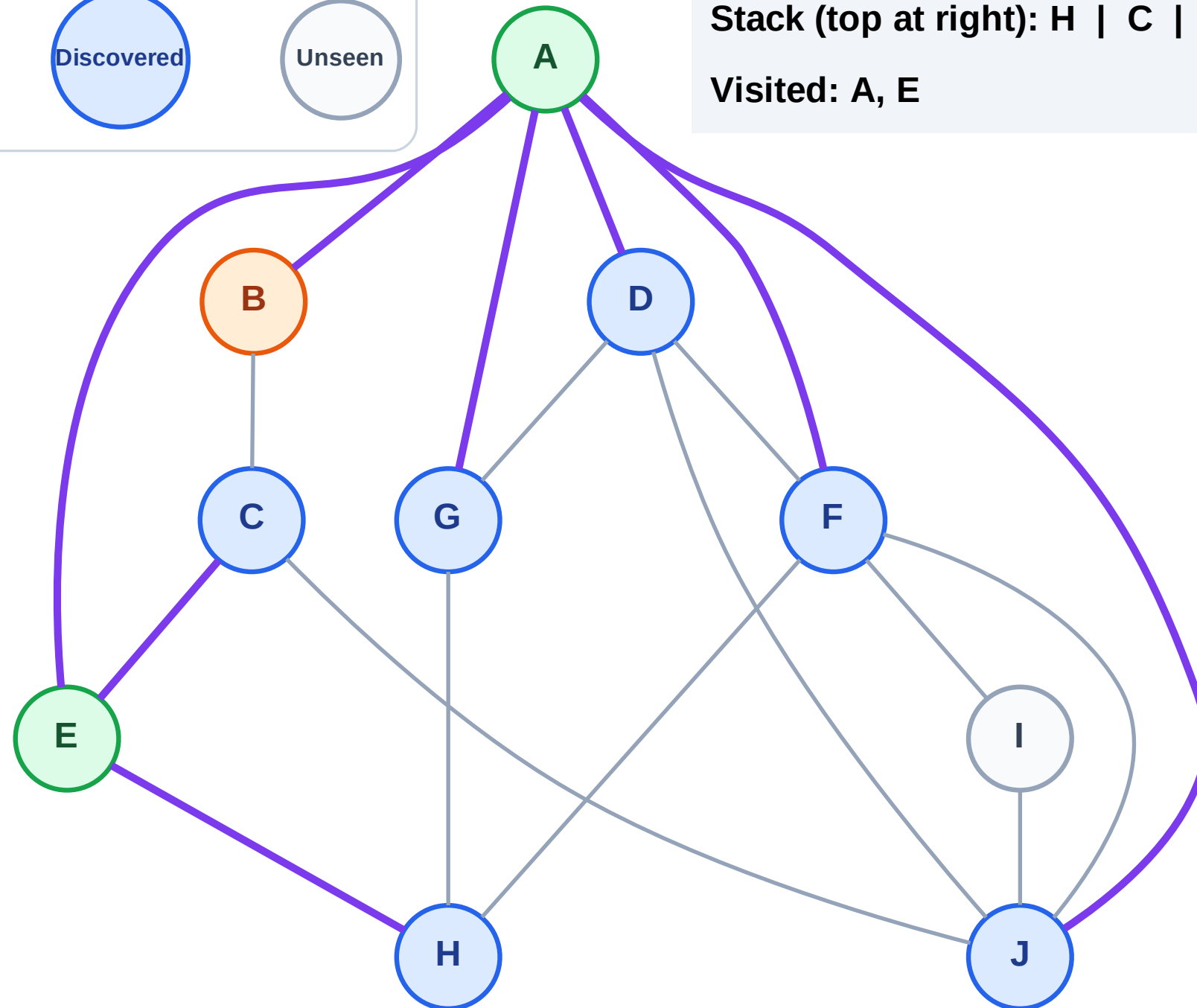
Step 6: Process node B

Legend



Stack (top at right): H | C | J | G | F | D

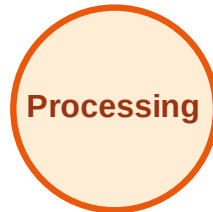
Visited: A, E



DFS Traversal

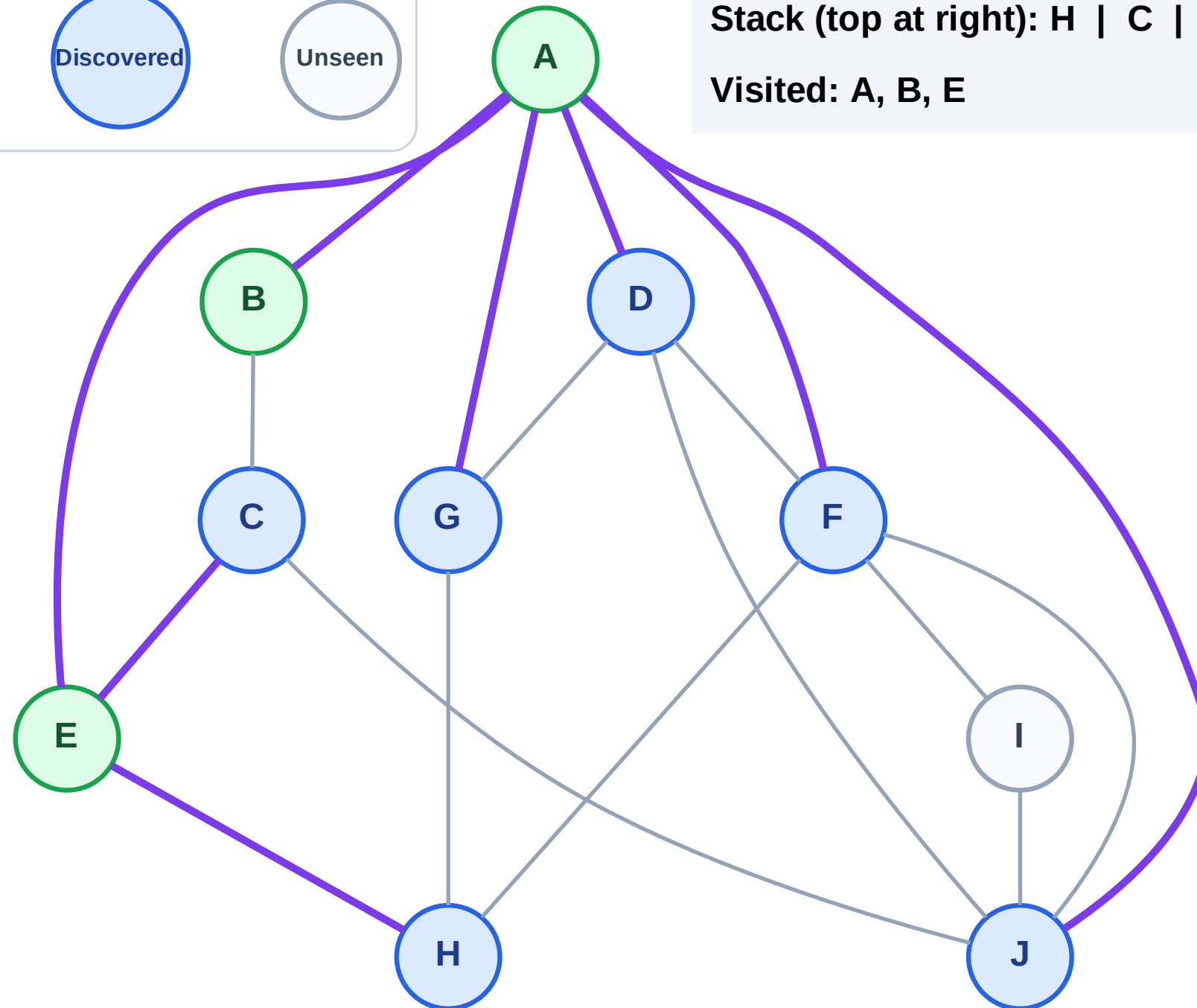
Step 7: No new neighbors from B

Legend



Stack (top at right): H | C | J | G | F | D

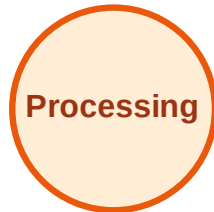
Visited: A, B, E



DFS Traversal

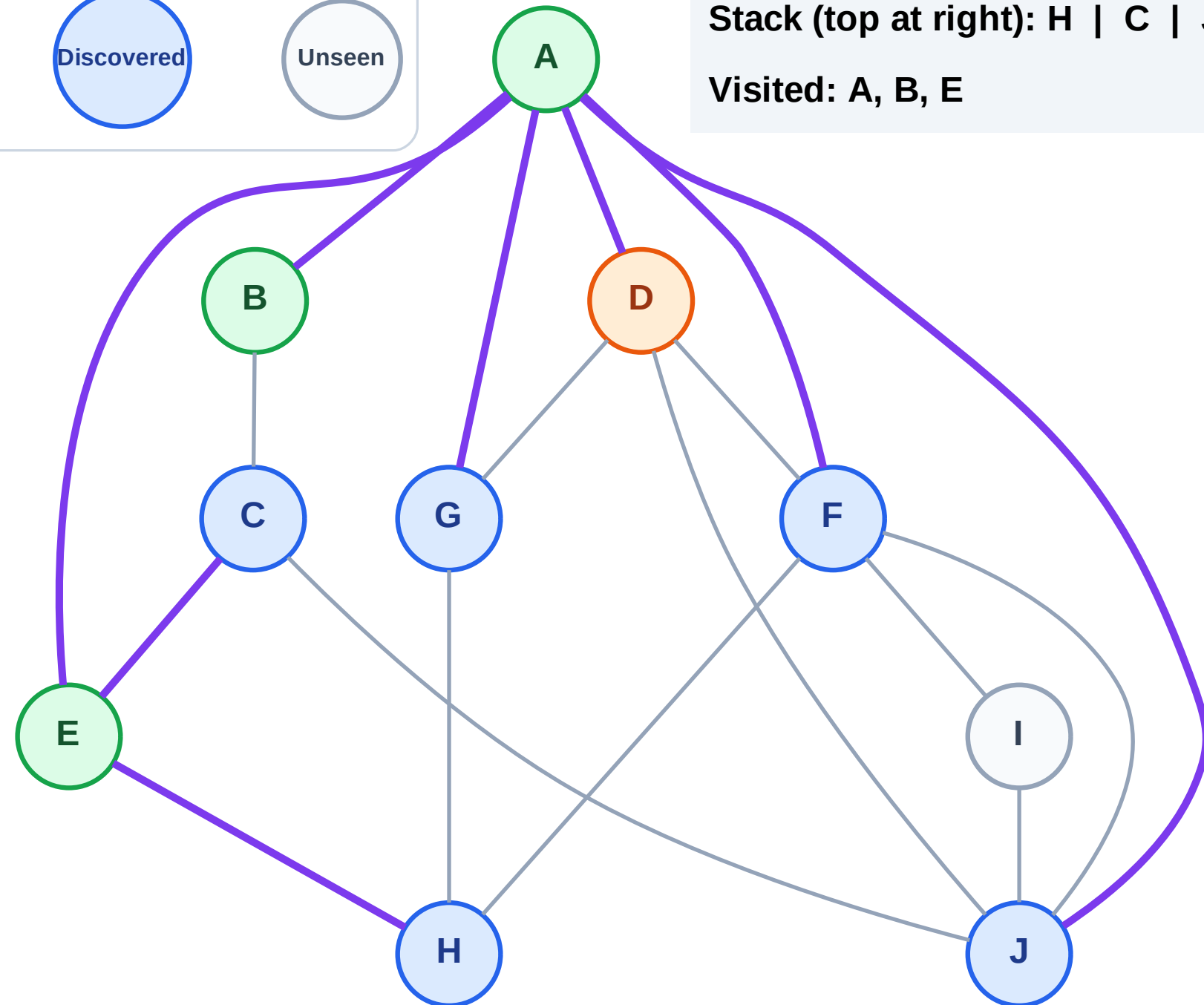
Step 8: Process node D

Legend



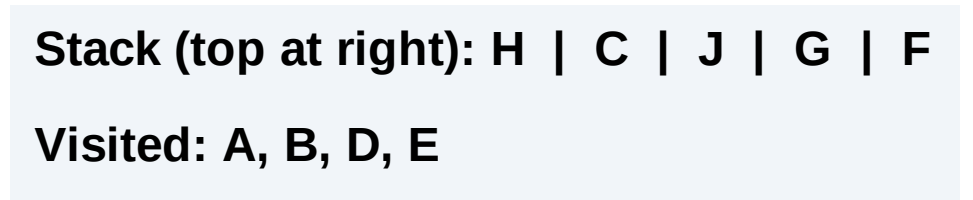
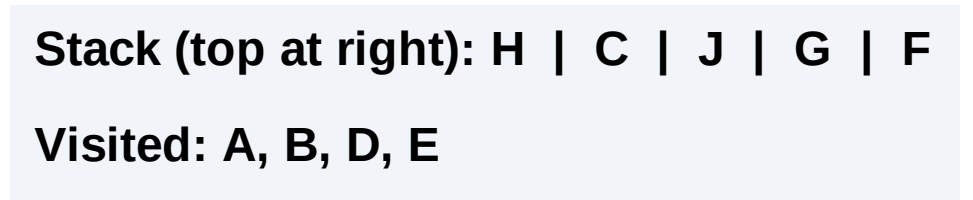
Stack (top at right): H | C | J | G | F

Visited: A, B, E



Step 9: No new neighbors from D

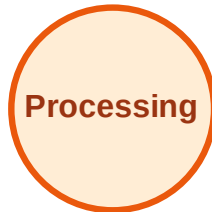
Step 9: No new neighbors from D



DFS Traversal

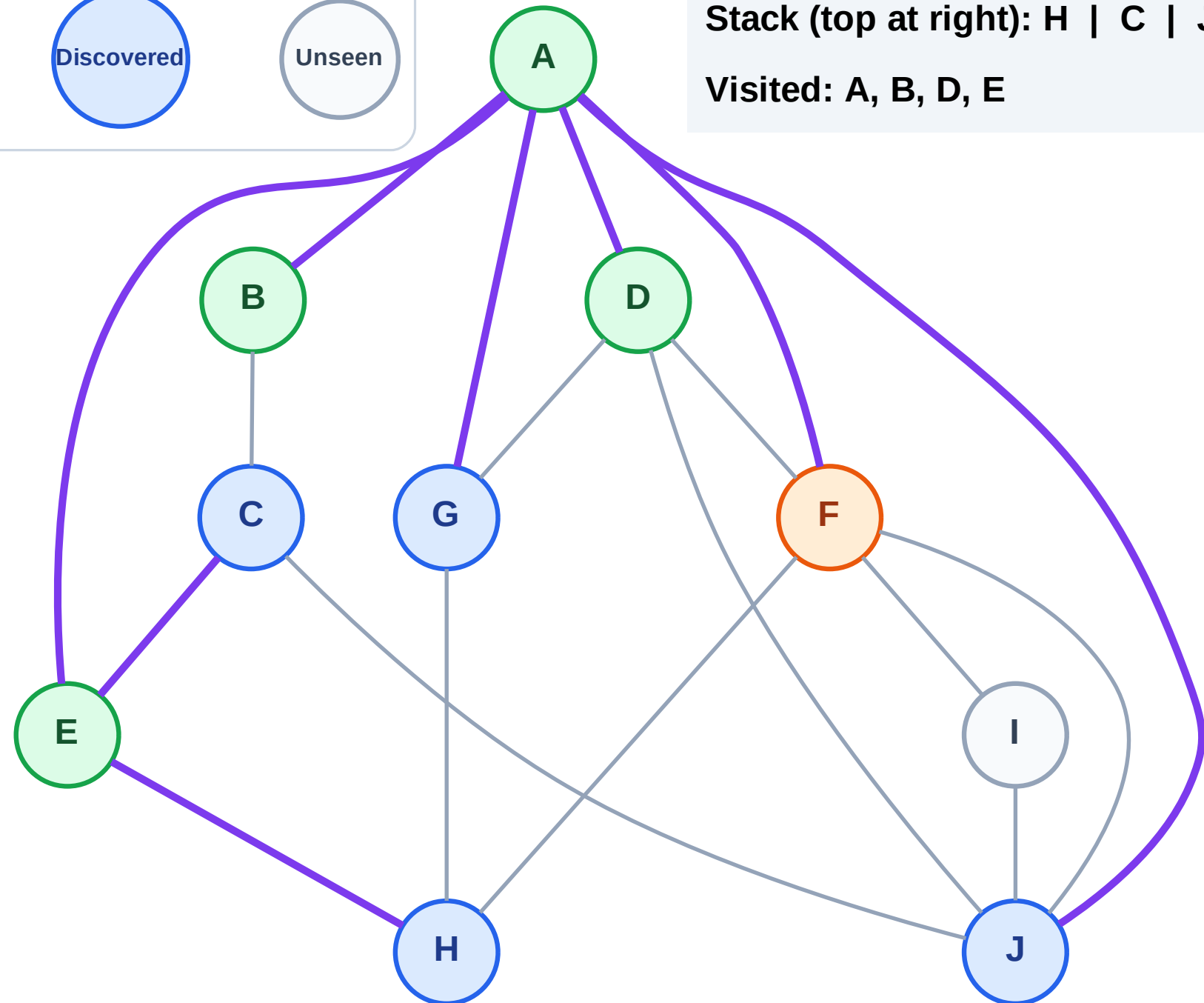
Step 10: Process node F

Legend



Stack (top at right): H | C | J | G

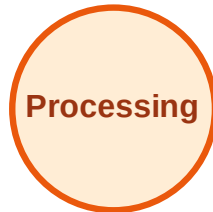
Visited: A, B, D, E



DFS Traversal

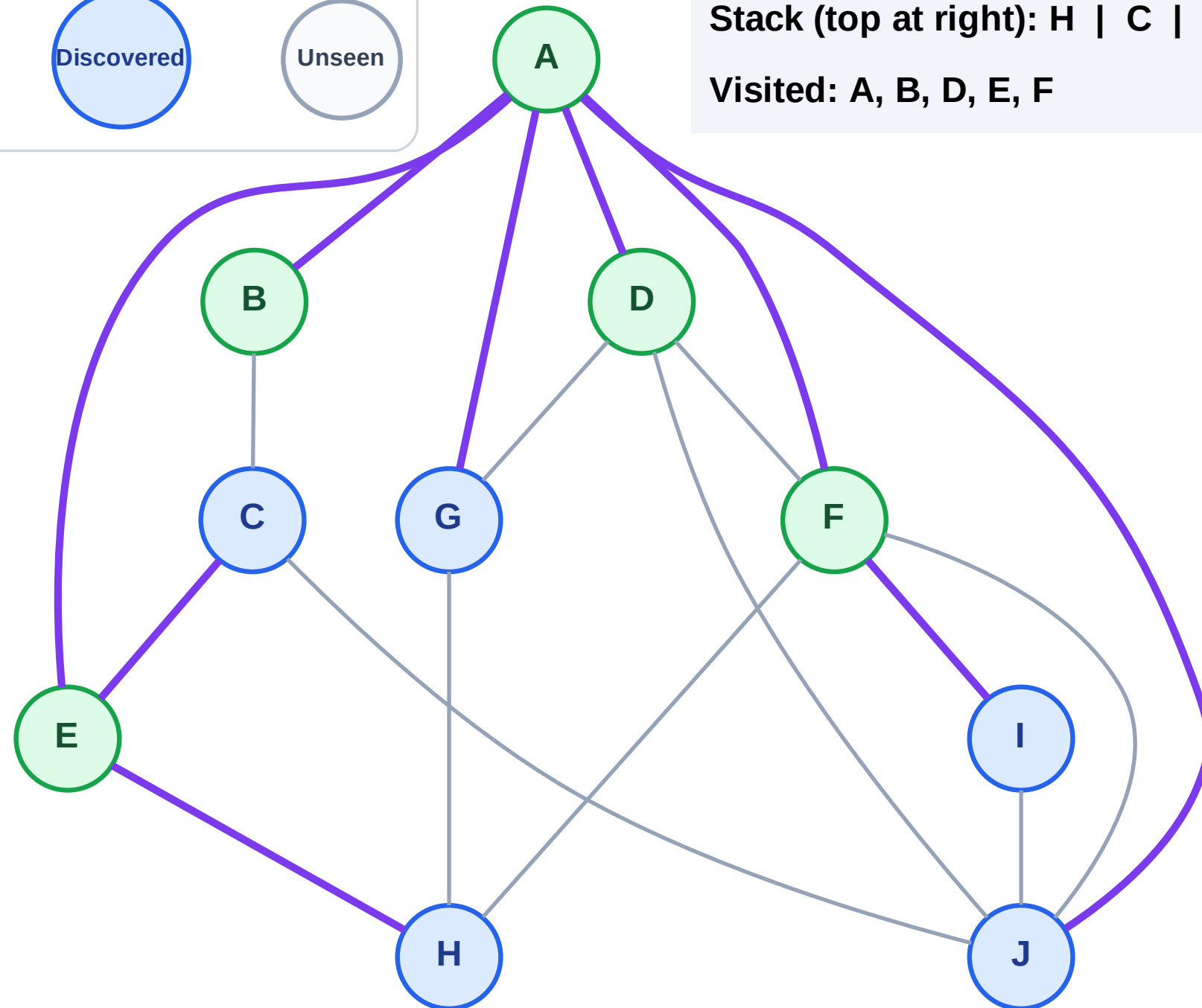
Step 11: Discover I from F

Legend



Stack (top at right): H | C | J | G | I

Visited: A, B, D, E, F



DFS Traversal

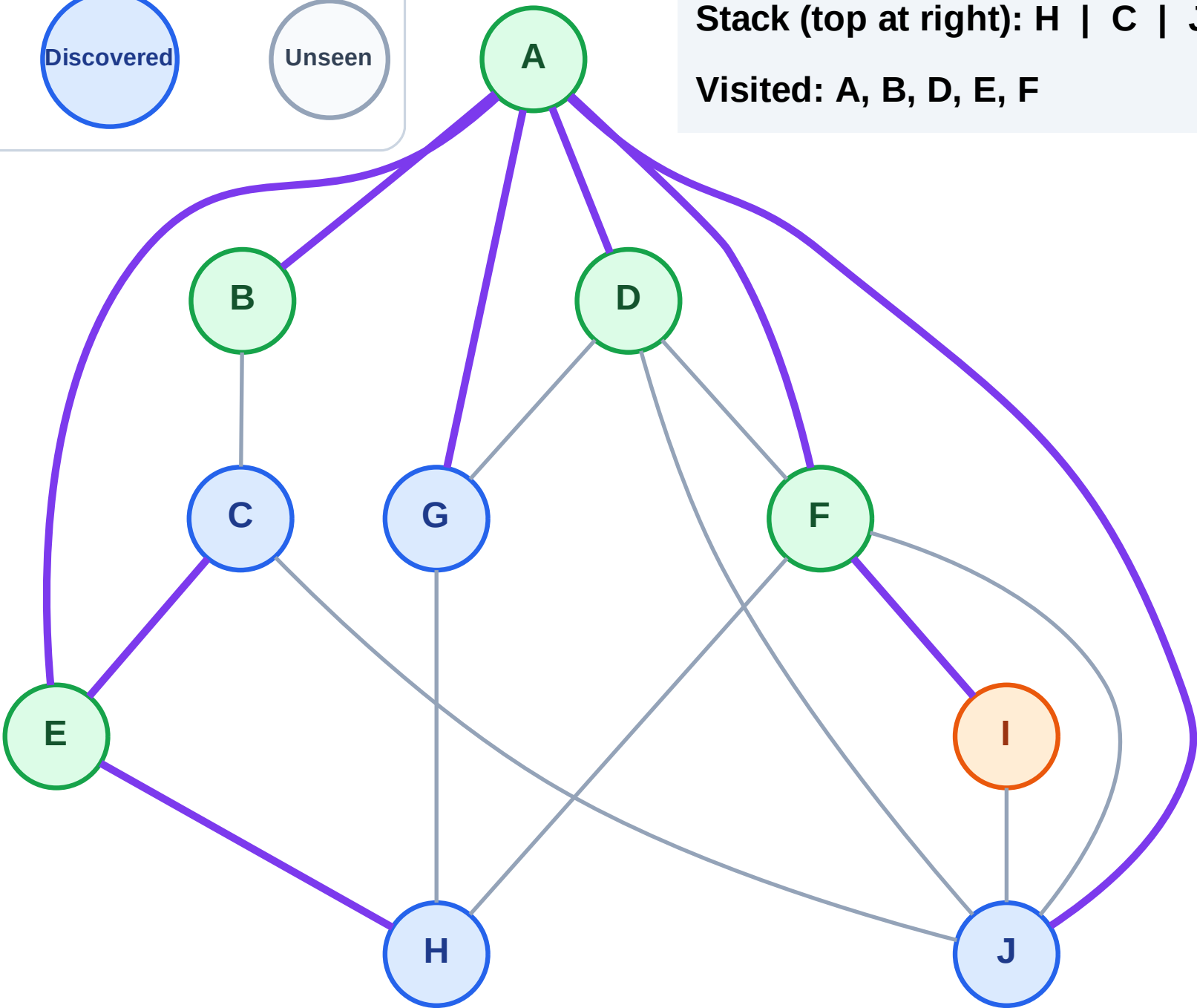
Step 12: Process node I

Legend



Stack (top at right): H | C | J | G

Visited: A, B, D, E, F



DFS Traversal

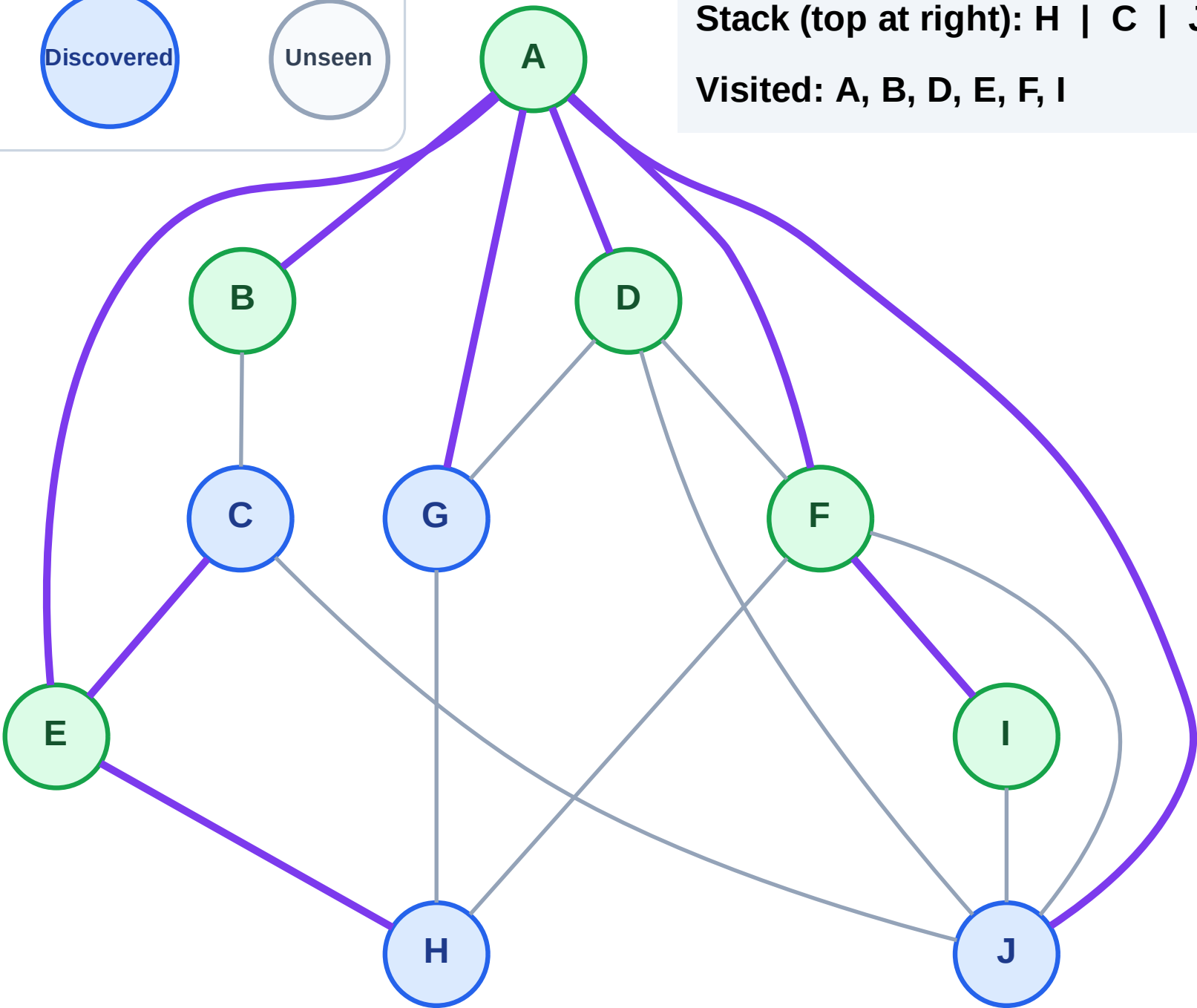
Step 13: No new neighbors from I

Legend



Stack (top at right): H | C | J | G

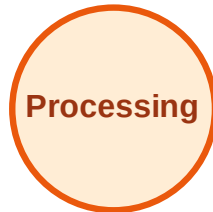
Visited: A, B, D, E, F, I



DFS Traversal

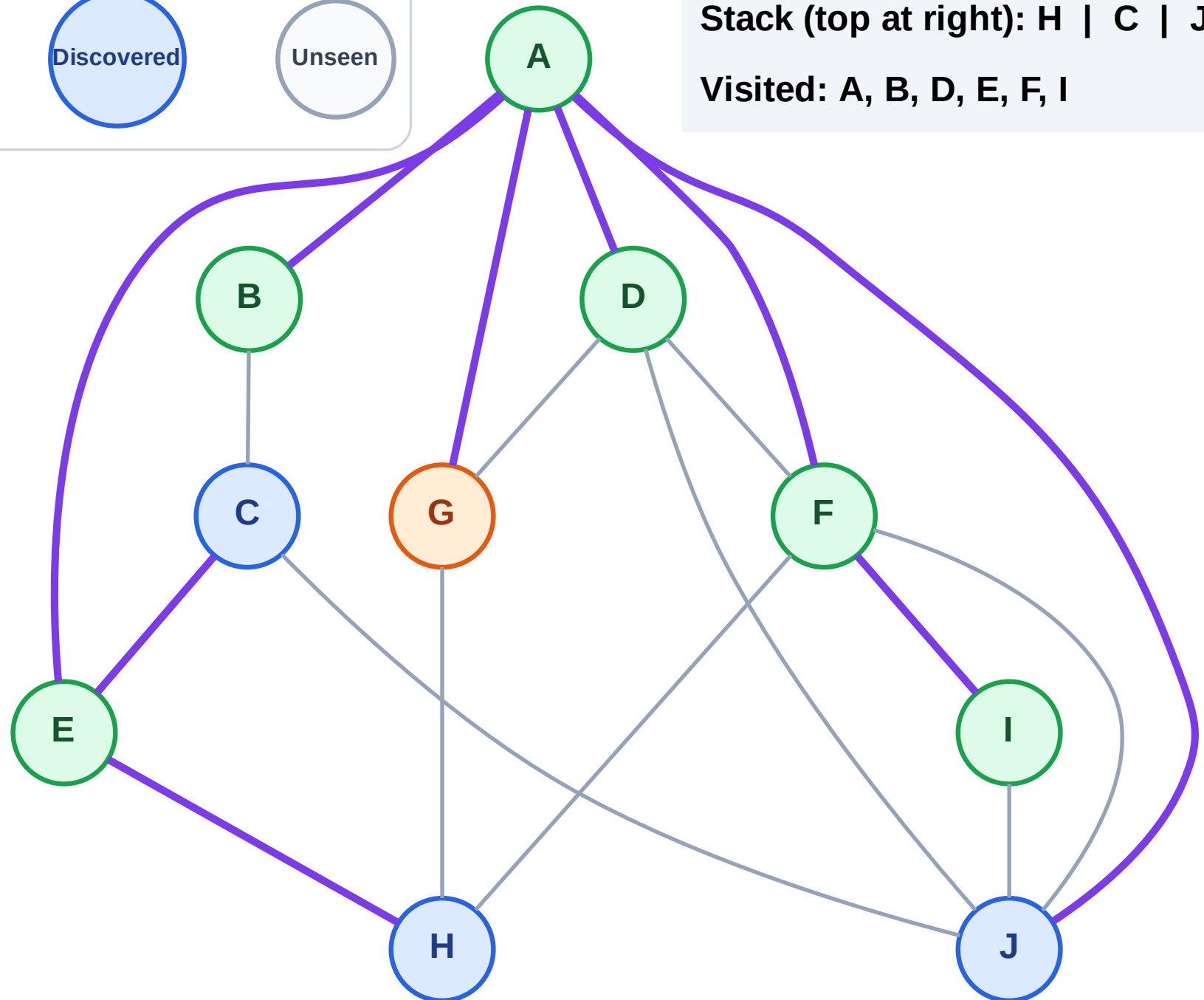
Step 14: Process node G

Legend



Stack (top at right): H | C | J

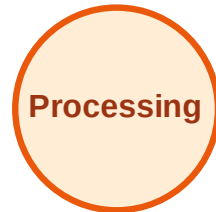
Visited: A, B, D, E, F, I



DFS Traversal

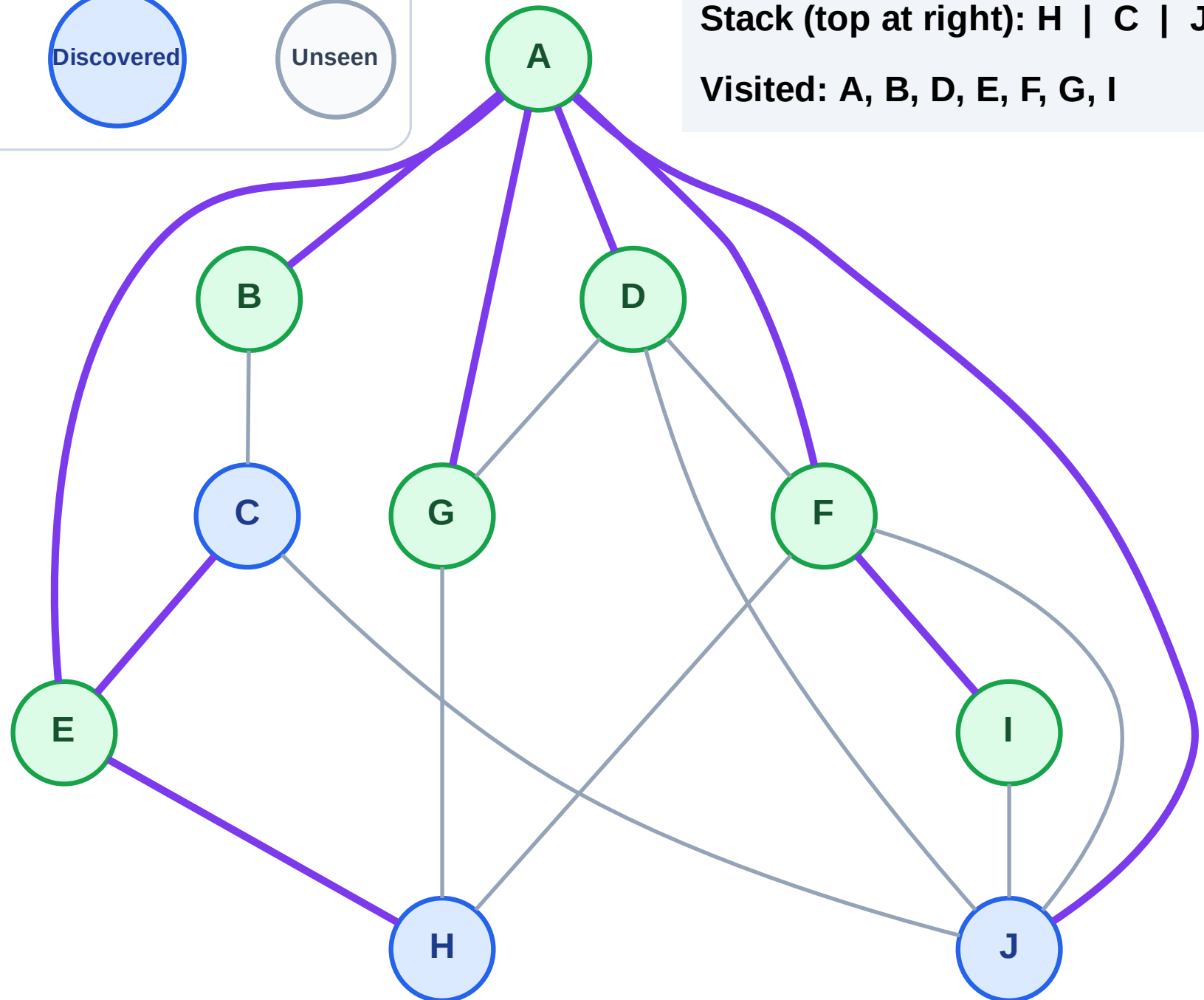
Step 15: No new neighbors from G

Legend



Stack (top at right): H | C | J

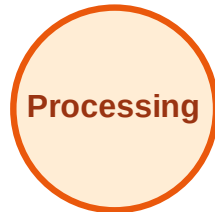
Visited: A, B, D, E, F, G, I



DFS Traversal

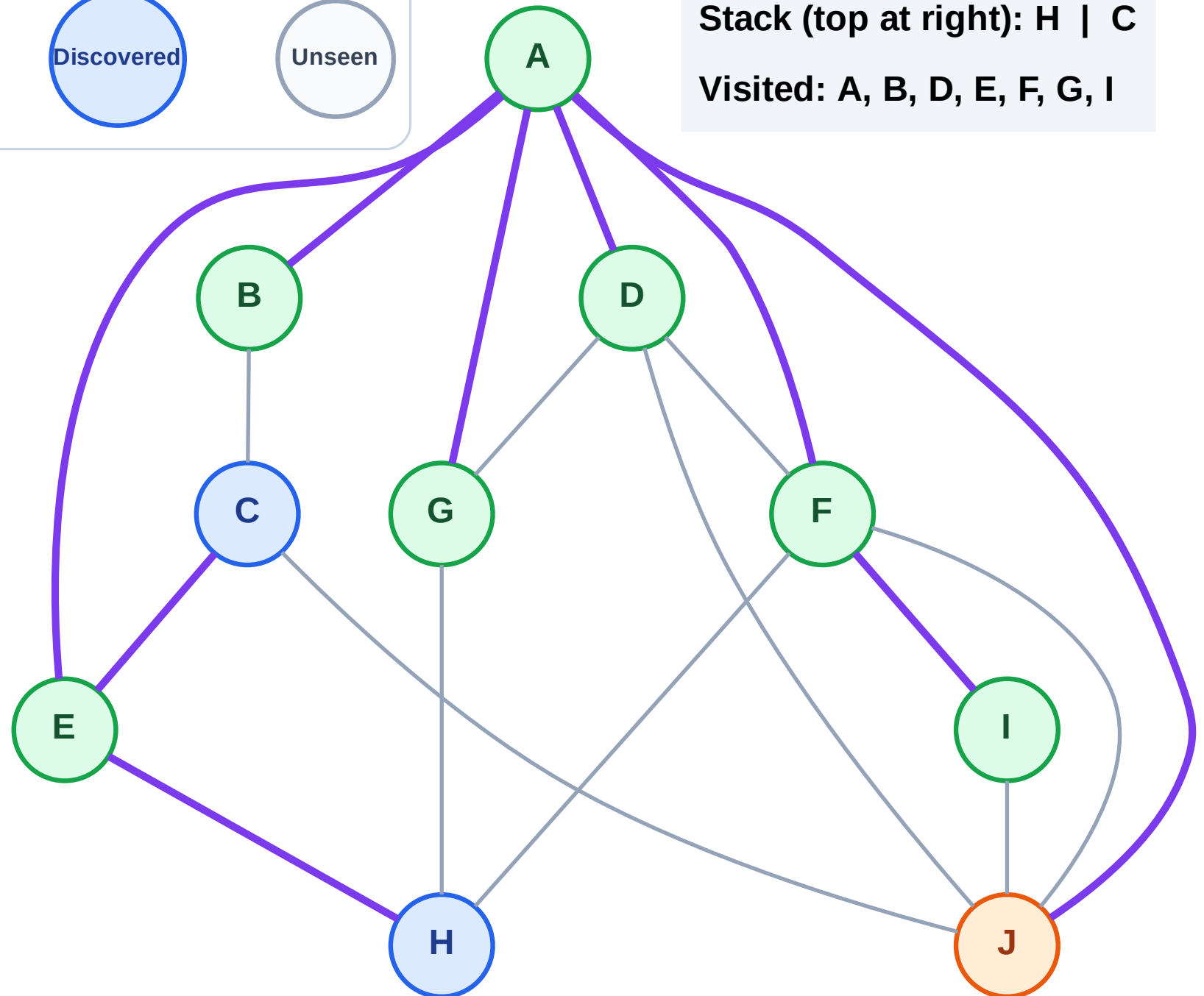
Step 16: Process node J

Legend



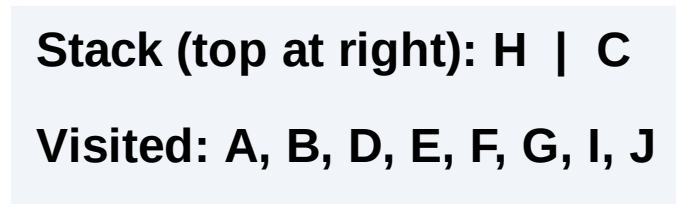
Stack (top at right): H | C

Visited: A, B, D, E, F, G, I



Step 17: No new neighbors from J

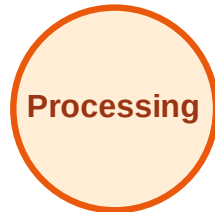
Step 17: No new neighbors from J



DFS Traversal

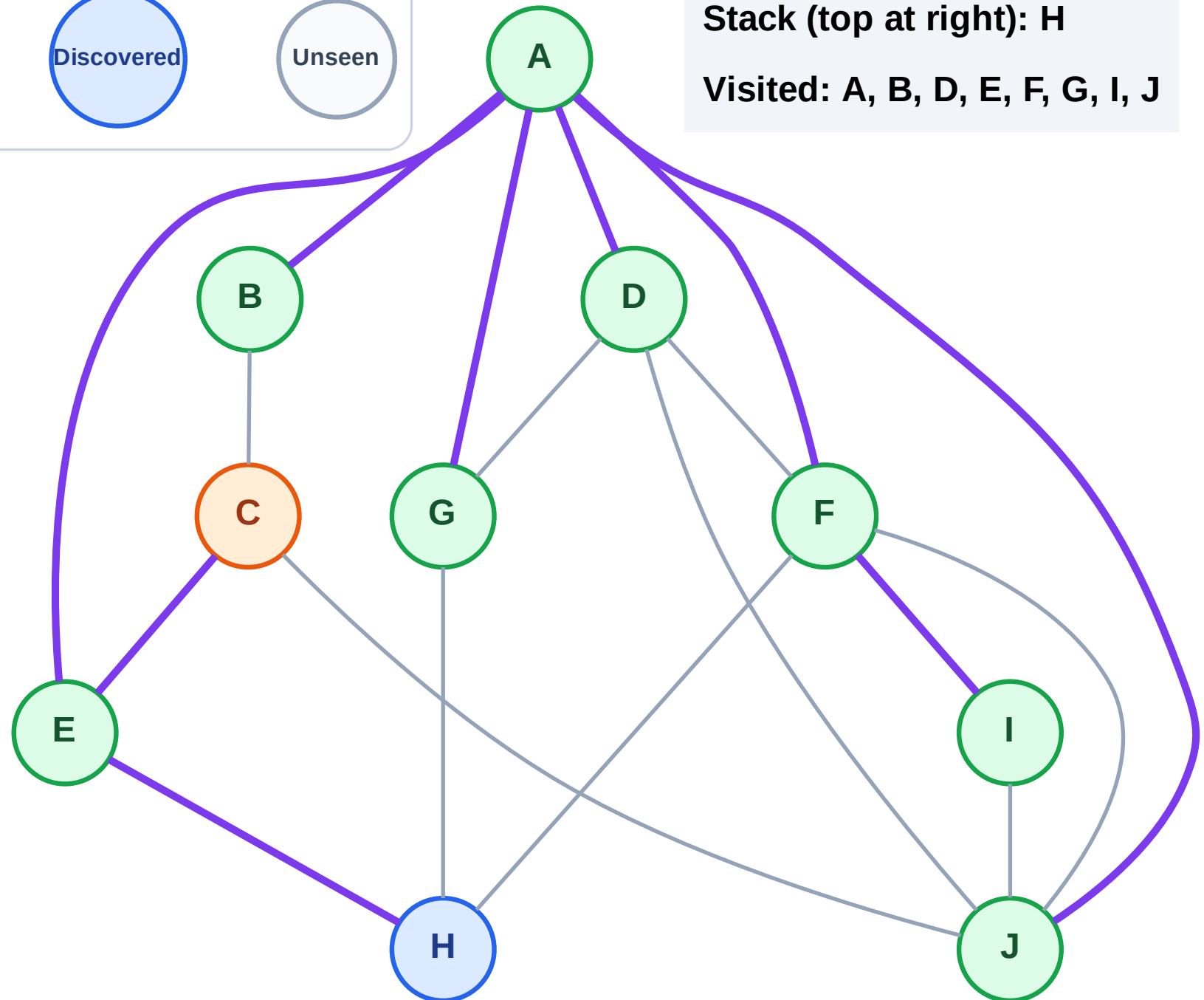
Step 18: Process node C

Legend



Stack (top at right): H

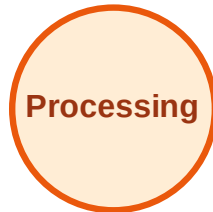
Visited: A, B, D, E, F, G, I, J



DFS Traversal

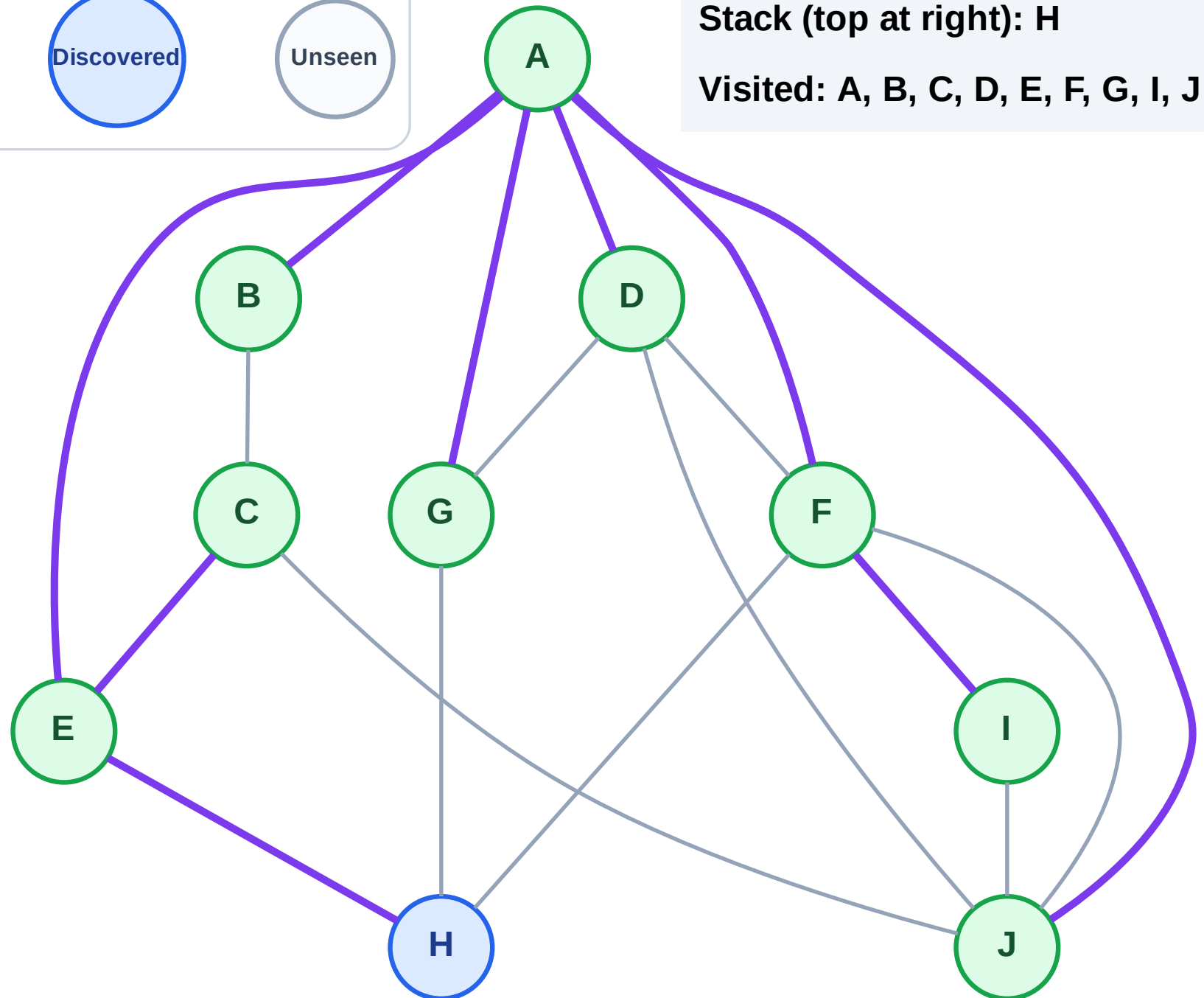
Step 19: No new neighbors from C

Legend



Stack (top at right): H

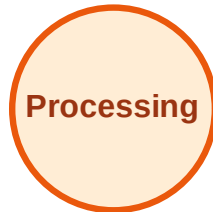
Visited: A, B, C, D, E, F, G, I, J



DFS Traversal

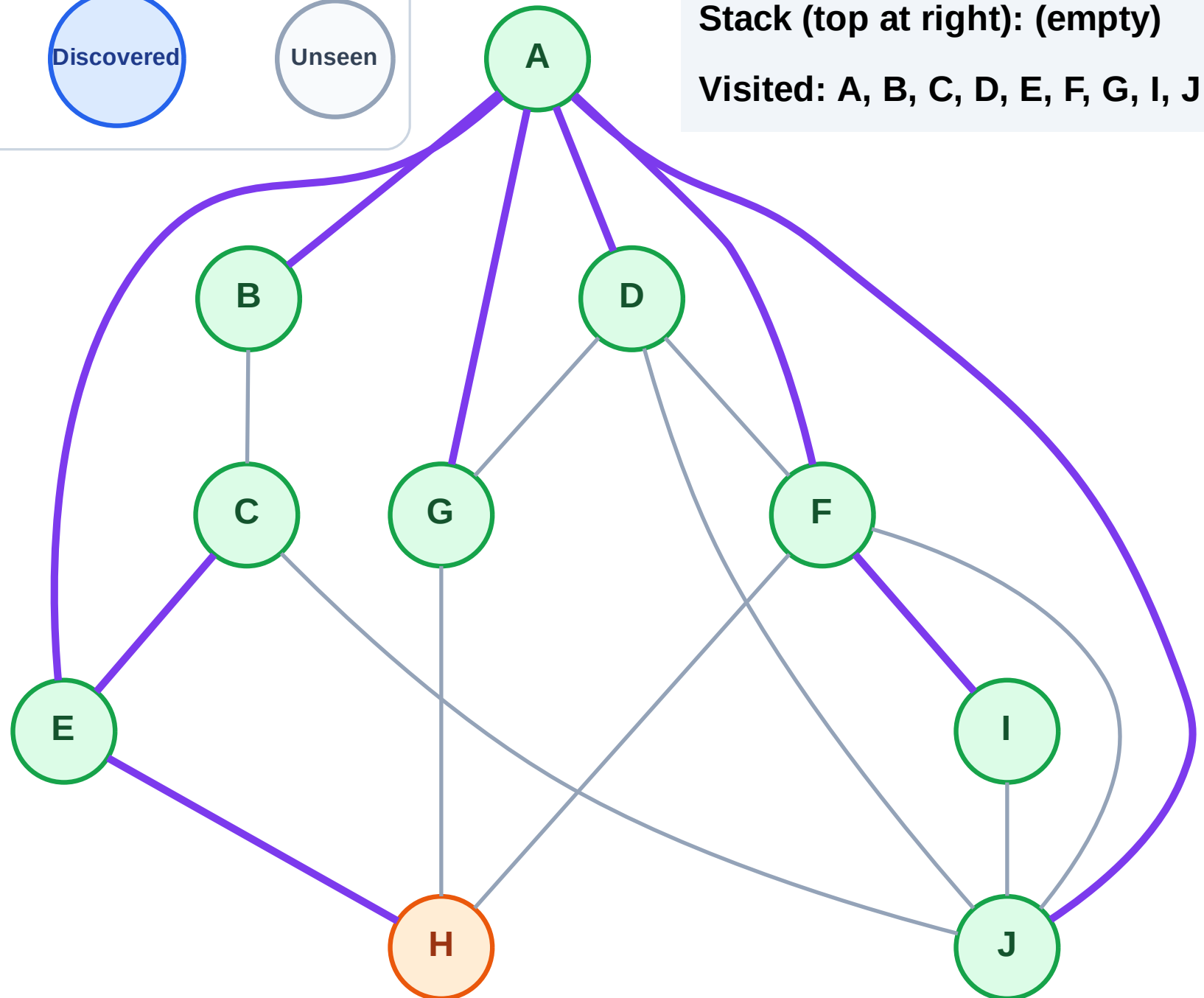
Step 20: Process node H

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, I, J



DFS Traversal

Step 21: No new neighbors from H

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

